CCAR-P · DOMAIN 3 · INTEGRATION · REPAIR WEEK

The Dock Gym

Eight bays, one error gym, one timed slice

104 exercises: 68 station items · 4 worked interrogations · 12 error-gym items · 8 tell hunts · 12-item form slice
Every station has warm-ups, exam items and head-to-heads; the four new objectives each have an autopsy



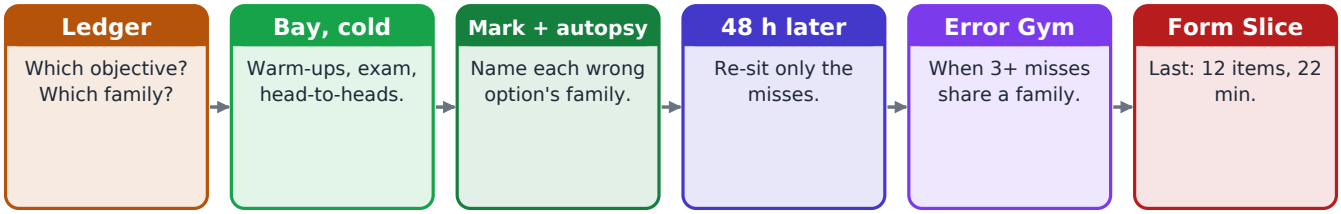
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Built 13 Oct 2026 · facts verified against the live docs 13 Oct 2026

THE LEDGER POINTS; THE GYM REPAIRS

How to use the Gym

This is the Domain 3 repair bench. You do not read it front to back. Your miss ledger (from Dock Book II, the D3 MR Drill and the mixed mocks) names the objectives you miss; you open those bays, sit them cold, mark them, and re-sit the misses 48 hours later.



The readiness gate · Domain 3 is ready when every row is ticked

Gate	Bar	Book	Done ✓
Dock Book II (45 q, cold)	≥ 34, no objective under 60%	existing D3 bench	
The Wrong Tower (30 flaws)	≥ 24 found, ≤ 2 false alarms	existing D3 bench	
D3 MR Drill (30)	≥ 24, OVER + UNDER ≤ 4, no objective with 3+ misses	new	
Gym bays 3.1–3.8	≥ 80% each, twice in a row on different days	this book	
Error Gym (12)	≥ 10	this book	
Tell hunt (8)	≥ 7 tells underlined exactly	this book	
Form Slice (12 in 22 min)	≥ 10 inside the time	this book	
Recall Deck pass 2 (40)	≥ 36	new	
Atlas A1–A7 redrawn	every box present	The Loading Dock	

Part	What	Items	Time
Bay 3.1	Evaluate tool/agent configuration for capability bloat	7	16 min
Bay 3.2	Analyze authentication and authorization requirements; identify security gaps	10	22 min
Bay 3.3	Evaluate accuracy–latency trade-offs and justify configuration decisions	7	16 min
Bay 3.4	Analyze observability challenges; select monitoring strategies at scale	10	22 min
Bay 3.5	Design a RAG pipeline with appropriate chunking and indexing strategies	10	22 min
Bay 3.6	Apply retrieval strategies matched to data shape and query pattern	10	22 min
Bay 3.7	Evaluate connection protocols; select the integration mechanism	7	16 min
Bay 3.8	Evaluate progressive discovery vs. monolithic context strategy	7	16 min
Interrogations	Holmes solves four aloud: code, a trace, an architecture, a bill	4	8 min each
Error Gym	12 items sorted by miss family: OP · OE · VG	12	25 min
Tell hunt	8 stems, no options: find the deciding clause	8	15 min
Form Slice	D3's share of a real form, timed	12	22 min
Keys	Station keys, error-gym key, tell-hunt answers, slice key		

THE FIVE LAWS AND THE NUMBERS THE EXAM PUNISHES

D3 quick card

Law	Domain 3 in one breath
1	**Every tool you add is a door; know who holds the key.**
2	**Chunk along meaning, not length.**
3	**Retrieval follows the shape of the data, not the fashion.**
4	**You cannot fix what you cannot see.**
5	**Send the map first, the territory on demand.**

Number	What it is	Objective
2,314 → 811	nine tool schemas vs the three used; saves 1,503 tokens a call = 60,120,000 tokens a day at 40,000 calls	3.1
310 ms vs 2,900 ms	mean vs p95 at the dock; one slow ticket's trace = 4,100 ms — the mean hid the tail	3.4
5.7 → 3.7 → 2.9 → 1.9%	top-20 retrieval failure: contextual embeddings (−35%), + contextual BM25 (−49%), + rerank (−67%)	3.5
~150 → rerank → 20	retrieve broadly, rerank, pass the top 20	3.5
< ~200k tokens	skip RAG: put the whole knowledge base in the prompt and cache it	3.5
362,460 vs 4,205 ≈ 86×	monolithic vs progressive per call (E8); ≈ 4,500-token small, stable set: send whole and cache	3.8
512 · 1,024 · 2,048 · 4,096	minimum cacheable prefix by tier (5.5 and Fable 5.1 → 512; Haiku 4.5 → 4,096); below it, nothing caches	3.8
1.25× / 2× · 0.1×	cache writes (5-min / 1-hour); reads 0.1× on most models (0.05× Opus 5.5 and Sonnet 5.5, 0.025× Fable 5.1); ≤ 4 breakpoints	3.8
~100k · keep 3	clear_tool_uses default trigger and the tool uses it keeps	3.8
stdio · Streamable HTTP	the two MCP transports; HTTP+SSE deprecated; tools · resources · prompts	3.7
₹6	ceiling per resolved exception (Devika) — every dial answers to it	3.3

Same word, two worlds. token (a credential vs a unit of text) · **index** (a book's back pages vs a vector store) · **prompt** (an MCP primitive the user invokes vs the system prompt) · **resource** (MCP data the app provides vs 'resources') · **scope** (an OAuth permission vs a code block) · **span** (one timed step in a trace) · **cache** (prompt cache vs a CDN) · **authN vs authZ** · **mean vs p95**.

OBJECTIVE 3.1

Bay 3.1 · Capability bloat

Evaluate tool/agent configuration for capability bloat

What this bay drills. Every tool schema costs tokens on every call, and lookalike tools hand the model the wrong wrench. Trim by usage logs AND by consequence: a rare tool that prevents harm stays.
Law 1: *Every tool you add is a door; know who holds the key.*
7 items · 2 warm-up · 3 exam · 2 head-to-head · about 16 minutes · bar ≥ 6 / 7

1 WARM-UP

Tern Logistics in Rotterdam runs a berth-planning agent that sends 16 tool schemas, 3,100 tokens in all, on every call. A month of usage logs shows only four tools ever called, and their schemas total 900 tokens. The agent makes 25,000 calls a day. If the team trims the catalogue to those four tools, how many input tokens a day does the trim save?

(A) About 77,500,000 tokens a day: the full 3,100-token catalogue, sent on each of the 25,000 daily calls

(B) About 22,500,000 tokens a day: the 900 tokens of the four kept schemas, sent on each of 25,000 calls

(C) About 55,000,000 tokens a day: the 2,200 trimmed schema tokens, no longer sent on each of 25,000 daily calls

(D) About 2,200 tokens a day, since tool schemas are read once when the agent starts and are then held in memory

2 WARM-UP

Sutlej Auto Parts' dealer-support agent carries two tools built by different teams: `get_part_price`, which reads the live catalogue, and `lookup_part_cost`, which returns the same list price from a nightly copy. Usage logs show the agent picks the nightly copy on about one call in nine, and dealers are quoted yesterday's price. What is the best configuration change?

(A) Keep both tools and re-run the lookup with the other tool whenever a dealer disputes the price that was quoted

(B) Add a third tool, `choose_price_source`, that the agent calls first to decide which of the two price tools to use on each call

(C) Merge the pair into one `get_part_price` tool on the live catalogue, with a description that says what it returns and from where

(D) Add a line to the system prompt telling the agent firmly to prefer `get_part_price` whenever a dealer asks for a price

3 **EXAM**

Palk Bay Fisheries runs a cold-chain agent with 11 tools whose schemas total 2,600 tokens; the platform team caps tool schemas at 1,500 tokens per call. Last month, five tools (1,150 tokens) handled every call but 14; five tools (1,240 tokens) were never called; and `escalate_to_shift_lead` (210 tokens) made those 14 calls, each time for a reefer-truck failure where a missed escalation spoils the catch on board. Which catalogue should the agent carry?

- **A** The five busy tools, plus a system-prompt line asking the agent to tell dispatch to phone the shift lead if a truck fails
- **B** The five busy tools plus `escalate_to_shift_lead`, 1,360 tokens in all, which fits the cap and keeps the costly rare path
- **C** The five busy tools only, 1,150 tokens, since a tool called 14 times a month earns less than its schema costs on every call
- **D** All 11 tools, 2,600 tokens, because any tool may be needed one day and a dropped tool causes failures nobody predicted

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4 **EXAM**

At Kaveri Freight, Arjun has trimmed KAVI's main agent from nine tool schemas (2,314 tokens a call) to the three it uses (811 tokens). He now adds a subagent that drafts customs-hold explanations: it reads shipment status and the customs SOP, then hands a draft back to KAVI, which replies to the customer. His first build passes the subagent the original nine-tool catalogue, `process_refund` included. What should the subagent receive?

- **A** The same three tools as KAVI's trimmed main agent, so that both agents share one tested configuration and one cached prefix
- **B** The full nine-tool catalogue, since KAVI already checks refund limits in code and the extra schemas cost only tokens
- **C** Only the read tools its drafting job needs, such as shipment status and SOP lookup, leaving refunds and replies with KAVI
- **D** The full catalogue, with a line in the subagent's prompt saying that it drafts text and must not call `process_refund`

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5 **EXAM**

Larkspur Health in Denver runs a patient-scheduling agent with 24 tools. A 90-day log review finds seven tools that were never called and that no safety path depends on; three lookalike search tools (`find_slot`, `search_availability`, `get_open_times`) that the agent confuses on about 6% of calls; and `escalate_to_nurse`, called 31 times, each for a patient describing urgent symptoms. Which two changes cut capability bloat without adding risk? Select TWO.

- **A** Split the agent into three subagents of eight tools each, so no single agent chooses among 24 schemas
- **B** Keep all 24 tools and add a prompt section explaining when each search tool should be preferred over the others
- **C** Also remove `escalate_to_nurse`, since 31 calls in 90 days is too rare to justify sending its schema on every call
- **D** Remove the seven tools that were never called in 90 days and that no safety or escalation path depends on
- **E** Merge the three search tools into one availability tool with a sharp description and clear, typed parameters

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6 **HEAD-TO-HEAD**

Tern Logistics' yard agent mixes up two tools. `hold_container` freezes a container in the yard pending customs; `cancel_booking` voids the shipment booking and releases the berth slot. Both are used every day, and their one-line descriptions read almost the same: 'stop a container from moving'. A reviewer proposes merging them into one tool to cut capability bloat. What should the team do?

- **A** Merge them into one `stop_container` tool with a `mode` parameter, which removes the lookalike pair from the catalogue
- **B** Keep both tools and add a prompt rule asking the agent to think carefully before it chooses between the two
- **C** Add a checking subagent that reviews every hold or cancel decision before the yard agent's tool call runs
- **D** Keep both tools and rewrite each description to say what it changes, when to use it, and how it differs from the other

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7 **HEAD-TO-HEAD**

Sutlej Auto Parts' warehouse agent answers 'is part 7310-K in stock at Ludhiana?' about 3,000 times a day. With no single-part tool, it calls `list_bin_contents` and reads the whole bin list, about 18,000 tokens, to find one row. A new `get_stock(part_no, site)` tool would add a 120-token schema to each of the agent's 9,000 daily calls. The agent also compiles the 06:00 replenishment report, which needs the full bin list. What should the team do?

(A) Keep the catalogue as it is: tell the agent in its prompt to stop reading the 18,000-token bin list once it finds the part

(B) Add `get_stock` and keep `list_bin_contents` so that the report still runs: about 1,080,000 schema tokens a day replace about 54,000,000

(C) Swap `list_bin_contents` out for `get_stock`, which fixes the 3,000 daily lookups and keeps the catalogue at its present number of tools

(D) Add `get_stock`, `search_parts`, `get_bin_history` and `get_site_summary`, so that every stock question has its own purpose-built tool

OBJECTIVE 3.2

Bay 3.2 · Authentication & authorization

Analyze authentication and authorization requirements; identify security gaps

What this bay drills. The gate pass (authentication: who are you?) is not the manifest of bays (authorization: what may you do?). Scopes checked in code before the tool runs; short-lived, per-integration keys; denials logged as well as grants.
Law 1: *Every tool you add is a door; know who holds the key.*
10 items · 3 warm-up · 5 exam · 2 head-to-head · about 22 minutes · bar ≥ 8 / 10

1 WARM-UP

Larkspur Health's remote MCP server checks each OAuth token's signature and expiry before it accepts a call. A security review finds that any valid token, including the one issued to the appointment-reminder bot, can call `export_patient_list`. Which kind of gap has the review found?

(A) An authorization gap: the server knows who is calling but does not check whether that caller's scopes allow this tool

(B) A transport gap: the server should move from Streamable HTTP to stdio, which narrows its scope to local processes only

(C) An authentication gap: the server should add multi-factor sign-in so that it is more certain who holds each token

(D) A token-budget gap: `export_patient_list` returns too many tokens for the model's context, so its output should be capped

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2 WARM-UP

A Palk Bay Fisheries analyst runs a small MCP server over stdio on her own laptop so that Claude Code can read her local folder of catch-landing spreadsheets. No one else connects to it. The security team asks where this server's authentication boundary sits. What is the best answer?

(A) At a session token on the HTTP+SSE connection that the server opens when Claude Code starts and holds until it exits

(B) At the laptop's operating-system login: the server runs as her local process and inherits her environment and file permissions

(C) At an OAuth access token: the MCP client requests one from the server before each tool call, as it would on any MCP transport, local or remote

(D) At an API key placed in Claude Code's system prompt, which the model passes to the server whenever it calls a tool

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3 WARM-UP

At Sutlej Auto Parts, a developer copied the dealer-chat agent's permanent order-system API key into a test script on his laptop. The laptop was stolen on Friday evening, and the first fraudulent call with that key came 40 hours later. Which credential design would most likely have made the copied credential useless by then?

- **A** A stronger 64-character permanent key from the secrets manager, long enough that it cannot be guessed by brute force
- **B** Short-lived OAuth access tokens that expire after 60 minutes, issued at run time to the agent's own service account
- **C** One shared key for all of the company's agents, held by the platform team, so that only one secret needs protecting
- **D** The same permanent key, rotated on a 90-day schedule by the secrets manager, which gives every key a known expiry date

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4 EXAM

Tern Logistics' booking agent in Rotterdam books berth slots with three carriers. To keep the integration simple, the developers placed all three carriers' API keys in the system prompt, and each booking tool takes an `api_key` argument that the model fills in. Last week the model sent the North Sea carrier's key to the Baltic carrier's endpoint, and a red-team test got the model to print a key in its reply. What is the best fix?

- **A** Encrypt the three keys inside the system prompt and give the model a `decrypt_key` tool that unlocks the right key for each booking
- **B** Keep the keys in the prompt but add a firm instruction not to reveal them and to match each key to its carrier's endpoint by name
- **C** Remove the keys from the prompt and tool arguments, and have each tool's handler load its own carrier's credential from a secrets store
- **D** Put each carrier behind its own subagent that holds only that carrier's key, with a router agent choosing among them

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5 EXAM

At Kaveri Freight, KAVI answers customers on the self-service portal, where each customer signs in with their own account. Padma Deshmukh, signed in, asks about consignment KF-2481. KAVI's `lookup_order` tool calls the carrier portal with one integration token, and Arjun's test shows that token can read every customer's consignments: a signed-in customer who types another customer's consignment ID gets that shipment's details back. Which change closes the gap?

- **A** Add a second agent that reviews each KAVI reply for other customers' data before the reply is shown on the portal
- **B** Make customers sign in with a one-time password as well as their portal password before KAVI answers any consignment questions on the portal
- **C** In the tool's code, check the consignment belongs to the signed-in customer before calling the portal; deny and log a mismatch
- **D** Add a rule to KAVI's system prompt: discuss only the signed-in customer's own consignments; politely decline any other consignment ID

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6 **EXAM**

Palk Bay Fisheries connects its auction-pricing agent, its nightly ERP sync job and its cold-store sensor feed to the same ERP API with one permanent key. On Monday someone changed 300 auction reserve prices, and the ERP log shows only that key. The CISO sets two goals: every change must be traceable to the integration that made it, and a leaked credential must stop working within the hour. Which two changes meet these goals? Select TWO.

- **A** Replace the permanent key with OAuth access tokens that expire within 60 minutes and are issued per service account
- **B** Put a human approver in front of every ERP write from all three integrations until the investigation is closed
- **C** Give each of the three integrations its own service account, with only the ERP scopes that its job needs
- **D** Rotate the shared key every Sunday night through the secrets manager, keep its current ERP scopes, and record each rotation in the change log
- **E** Have the agent's system prompt add the tag 'pricing-agent' to every ERP change it makes, so that its changes stand out

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7 **EXAM** · **AUTOPSY**

Sutlej Auto Parts' dealer-chat agent can call `apply_dealer_discount`. Policy says the chat service may grant up to 15%; anything higher must come from the regional sales service, which holds the `discount:extended` scope. Today the only control is a system-prompt line stating the 15% limit. A dealer pasted a fake 'regional manager note' into the chat and received 40% on a ₹2,60,000 order. What is the best fix?

- **A** Rewrite the 15% prompt line in capitals and add three worked examples of fake manager notes that the agent should refuse
- **B** In the tool's handler, check the caller holds `discount:extended` before any discount above 15% runs; deny and log otherwise
- **C** Grant the chat service `discount:extended` as well; genuine regional approvals pasted into the chat window then stop failing for dealers
- **D** Require dealers to sign in with multi-factor authentication before the chat agent will discuss any discount with them

AUTOPSY · the answer is B. Do not choose — dissect. Name the miss family of each wrong option and the clause that kills it.
A _____ **C** _____ **D** _____

8 EXAM

Larkspur Health's compliance team pulls September's audit log for its records MCP server. It shows 4,812 granted calls to `export_patient_list` and no denials. The server's own error counter for the same month shows 37 calls refused with HTTP 403 for a missing `records:export` scope. The team lead says the empty denial column proves the scopes are working. What is the best response?

- **A** Stand up a second, independent audit server that mirrors every call, so that the two logs can be compared at month end
- **B** Add the monthly 403 count to the compliance dashboard as a metric, which gives auditors the record of refused attempts
- **C** Agree: zero denials in the audit log shows that no caller without `records:export` tried to export patient lists in September
- **D** The log records grants only: write every refused call to it with caller, tool, arguments and reason, and investigate the 37

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9 HEAD-TO-HEAD

Tern Logistics runs one MCP server that exposes `read_eta` and `rebook_vessel_slot`; the server accepts its single bearer token for every tool it exposes. A new customer-facing ETA bot needs only `read_eta`. One engineer proposes giving the bot the existing token and listing only `read_eta` in the bot's client configuration. Another proposes a separate token for the bot. What should the team do?

- **A** Issue the bot its own token scoped to `eta:read`, and have the server check that scope in code before any tool runs
- **B** Give the bot the existing token and state in its system prompt that rebooking is outside its role with customers
- **C** Clone the server into a second deployment that exposes only `read_eta`, with its own token, hosting and upkeep
- **D** Give the bot the existing token and filter its client tool list in code to `read_eta`, which scopes the bot to one tool and trims its schemas

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10 HEAD-TO-HEAD

Larkspur Health's scheduling MCP server began as a stdio prototype on one developer's machine. It exposes `book_appointment` and `cancel_appointment`, and it will now serve 12 front-desk agents across Denver clinics from one hosted copy. One architect wants to keep stdio, because it is simple and needs no sign-in. Another wants Streamable HTTP with OAuth. Which design should the team choose?

- **A** Host it over Streamable HTTP behind one shared API key, installed by IT on all 12 front-desk computers, with tool scopes checked in code
- **B** Keep stdio and install a copy on each front-desk computer, so that every copy inherits that clerk's operating-system login
- **C** Host it over HTTP+SSE with a session token per clinic, the remote transport that most MCP tutorials still describe
- **D** Host it over Streamable HTTP with per-client OAuth tokens, and check each tool's scope in code before it runs

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A WORKED SOLUTION · STIMULUS: CODE

Interrogation IR3-1 · bay 3.2

Try it yourself first: 4 minutes. Then fold and read Holmes solving it aloud.

1 Larkspur Health in Denver hosts its billing MCP server over Streamable HTTP. Two clients hold valid OAuth tokens: the front-desk scheduling bot and the billing-office agent, and only the billing office's tokens carry the `billing:refund` scope. A penetration test shows the scheduling bot's token can refund any paid invoice. The exhibit is the tool handler. What is the best fix?

```
 1  @mcp.tool()
 2  def issue_patient_refund(invoice_id: str, amount: float) -> dict:
 3      claims = verify_jwt(current_request().bearer_token)  # signature + expiry
 4      log.info("refund_call", client=claims["client_id"], invoice=invoice_id)
 5      invoice = billing.get_invoice(invoice_id)
 6      if invoice is None:
 7          return {"error": "invoice not found"}
 8      if amount > invoice.paid_amount:
 9          return {"error": "amount exceeds amount paid"}
10      refund = billing.issue_refund(invoice, amount)
11      return {"status": "ok", "refund_id": refund.id}
```

(A) Add a line to the scheduling bot's system prompt saying refunds are outside its scope and it must not call this tool

(B) After line 3, check that the token's scopes include `billing:refund`, and if not, log the denial and return an error before line 10

(C) Remove `issue_patient_refund` from the scheduling bot's client tool list, which stops the bot receiving the tool's schema at all

(D) Extend `verify_jwt` on line 3 to check each token's audience and issuer as well, so that only genuine Larkspur tokens ever reach line 10

Write your letter and the clause that decided it here, before you turn the page:

_____

✂ HOLMES, ALOUD · READ AFTER YOU HAVE COMMITTED TO A LETTER ✂

The moves

1. Read once: a hosted server, two clients with valid tokens, and one tool that moves money. The pen test says the wrong client got through the door.
2. Underline the tell: 'only the billing office's tokens carry the `billing:refund` scope'. The scope exists. The question is whether any line of code ever reads it.
3. Walk the code: line **3** proves who is calling, which is authN. Lines **4** to **9** check the invoice and the amount. Nothing asks what this caller may do. The token's scopes are never read, and line **10** moves the money.
4. Name the objective: **3.2**, authN against authZ. The gate pass is checked; the list of bays it may enter is not.
5. Kill the vocabulary trap: a stricter `verify_jwt` (audience, issuer) is still authN. The scheduling bot's token is genuine; it would sail through.
6. Kill the sign and the half-lock: a prompt line is a sign, and dropping the tool from the bot's client list trims a schema, not a permission. Whoever holds the token still reaches line **10**.
7. Let the fixed requirement pick: the refund must be impossible without `billing:refund`. Only a scope check in the handler, before line **10,** does that, and writing the refusal to the log gives the auditors the attempt as well as the grant.

THE VERDICT

*The handler asks the visitor 'who are you?' and, satisfied, never asks 'and what may you do here?'. That is the whole crime, Watson. A token is a gate pass, not a manifest of bays: the scheduling bot's pass was perfectly genuine, which is why a stricter identity check would have changed nothing. Put the lock on the door it guards: read the scope after line **3,** refuse and log before line **10**. A prompt line or a shorter tool list leaves the door open to anyone who holds the key.*

1. **Answer B**

The tell only the billing office's tokens carry the `billing:refund` scope

B ✓ The handler proves who is calling but never reads what that caller may do; a scope check in code before `issue_refund` is the lock, and logging the denial gives auditors the attempt.

A **SD** A sign, not a lock; anyone holding the bot's token can still call the tool directly.

C **OP** A client-side filter trims schemas, not power; the bot's token still opens the refund tool on the server.

D **VG** More authN: the scheduling bot's token is genuine and would pass any stricter identity check. The gap is authZ.

↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

OBJECTIVE 3.3

Bay 3.3 · Accuracy–latency trade-offs

Evaluate accuracy–latency trade-offs and justify configuration decisions

What this bay drills. One sentence justifies a dial: constraint + dial + direction + number. When both options clear the constraint, the cheaper or faster one wins; when only one does, the constraint decides.
7 items · 2 warm-up · 3 exam · 2 head-to-head · about 16 minutes · bar ≥ 6 / 7

1 WARM-UP

Sutlej Auto Parts' architecture board accepts a configuration change only if its decision-log entry is one sentence naming the constraint, the dial that moved, the direction it moved and the number it cost or bought. Which entry meets the board's rule?

(A) We cut top-k on part lookups to make the assistant faster and more efficient, with no noticeable loss of answer quality

(B) To meet the 3 s p95 SLA on part lookups, we cut top-k from 30 to 15, which cost 0.4 points of golden-set accuracy

(C) We moved part lookups to a lower top-k, in line with the vendor's recommended defaults for retrieval assistants

(D) To meet the 3 s p95 SLA on part lookups, we cut the p95 from 3.4 s to 2.6 s, at a cost of 0.4 points of accuracy

2 WARM-UP

Larkspur Health's after-visit summary takes about 7 s to generate, and the text appears only when it is complete. Usability tests show patients leave if no words appear within 1 s; they happily read along once text is flowing. Accuracy is at the bar. A test with streaming shows the first words at 0.6 s. Which change meets the 1 s goal?

(A) Turn on streaming: the first words reach the screen at about 0.6 s, while the full summary still takes about 7 s

(B) Move the summary to the smallest model tier, which finishes in about 2.5 s and shows the complete text at once

(C) Turn on streaming: the whole summary then arrives in about 0.6 s, so the 7 s wait disappears for patients

(D) Send summaries through the Message Batches API, which halves their cost and frees capacity for live requests

3 EXAM

Tern Logistics' ETA agent answers a terminal planner by calling three tools one after another: vessel position (1.2 s), berth schedule (0.9 s) and customs status (1.5 s). None of the three needs another's result. Model time across the turn is 1.4 s, so the answer takes 5.0 s against an SLA of 4 s, and accuracy must stay at the current 95% on the ETA eval. Which change meets both?

(A) Stream the planner's answer, so that the first words appear within about a second and the 5.0 s wait no longer counts

(B) Move the agent to the smallest model tier, which trims model time and brings the whole turn to about 4.4 s

(C) Drop the customs-status call, the slowest at 1.5 s, and let the model infer customs status from the berth schedule

(D) Let the model request all three tools in one turn and run them in parallel, so the turn takes about 1.5 + 1.4 = 2.9 s

4 EXAM

Palk Bay Fisheries' export-rules assistant must answer with p95 ≤ 4 s and at least 92% on its golden set. This sprint the team cut top-k from 40 to 20 (p95 4.6 s → 3.7 s; accuracy 93.1% → 92.6%), then lowered effort from high to medium (p95 3.7 s → 3.2 s; accuracy unchanged at 92.6%). Which two sentences belong in the decision log? Select TWO.

[A] To keep headroom under the 4 s p95, we lowered effort from high to medium, saving 0.5 s at no accuracy cost (92.6% both ways)

[B] To meet the 4 s p95, we cut top-k from 40 to 20, taking p95 from 4.6 s to 3.7 s for 0.5 points of accuracy

[C] To meet the 4 s p95, we cut top-k from 40 to 20, which brought the average response time down to 2.1 s

[D] We reduced top-k and effort to make the assistant faster and cheaper, while keeping answer quality at a level users accept

[E] We cut top-k from 40 to 20 and set effort to medium, following the recommended defaults for retrieval assistants

5 EXAM

Larkspur Health's referral-letter classifier routes letters to clinics; a wrong route delays a patient's first appointment. The acceptance bar is ≥ 95% on the golden set with p95 ≤ 6 s. Test results: smallest tier 91.0% at 1.1 s p95; mid-tier 96.5% at 2.4 s; largest tier 97.5% at 5.2 s. The team has run on the smallest tier since the pilot because it is fastest. Which configuration should go to production?

(A) The largest tier: 97.5% routes the most letters correctly, and its 5.2 s p95 still fits inside the 6 s limit

(B) The smallest tier with streaming turned on, so that its letters are routed sooner and staff wait less for each one

(C) The mid-tier model: it is the cheapest tier that clears the 95% bar, and its 2.4 s p95 sits well inside 6 s

(D) The smallest tier: its 1.1 s p95 gives the best experience, and 91.0% is close enough to the 95% bar

6 HEAD-TO-HEAD

Tern Logistics' berth-status assistant has an acceptance bar of ≥ 92% on the golden set and p95 ≤ 2 s. Configuration P, the smallest tier with medium effort, scores 93.5% at 1.2 s p95. Configuration Q, the mid-tier with medium effort, scores 96.0% at 1.8 s and costs about four times as much per call. The service contract pays nothing for accuracy above the bar. Which configuration should ship?

(A) Configuration P with effort raised to high, to pull its accuracy up toward Q's 96.0% at the smaller tier's price

(B) Configuration P: both clear the bar, so the tie-break takes the cheaper and faster one, at about a quarter of the cost

(C) Configuration Q: both clear the bar, and 2.5 extra points of accuracy are worth having while the 2 s p95 still holds

(D) Configuration Q with streaming turned on, so that planners see its first words sooner and the slower tier is not felt

7 HEAD-TO-HEAD

Sutlej Auto Parts' order-validation agent returns a JSON verdict to the ERP, which can act only on the complete object. The integration SLA is p95 ≤ 3 s for the complete response; the agent now runs at high effort, 4.2 s and 97.9% accuracy against a 97% bar. Tests show medium effort gives 2.8 s and 97.6%, the smallest tier gives 1.9 s and 93.0%, and streaming brings the first token to 0.5 s. What should the team change?

(A) Send validations through the Message Batches API, which halves their cost and frees capacity for the remaining live calls

(B) Lower effort from high to medium: the complete verdict arrives at 2.8 s p95, and 97.6% still clears the 97% bar

(C) Turn on streaming: the first token arrives at 0.5 s, which brings the agent well inside its 3 s SLA at no accuracy cost

(D) Move to the smallest tier: the verdict arrives at 1.9 s p95, which leaves the most headroom under the 3 s SLA

OBJECTIVE 3.4

Bay 3.4 · Observability at scale

Analyze observability challenges; select monitoring strategies at scale

What this bay drills. Logs say what happened, traces say where the time went for one request, metrics say how often and how slow. Promises live in the tail (p95, p99), never in the mean. IDs belong in traces, not metric labels.
Law 4: *You cannot fix what you cannot see.*
10 items · 2 warm-up · 5 exam · 3 head-to-head · about 22 minutes · bar ≥ 8 / 10

1 WARM-UP

Palk Bay Fisheries' auction-quote assistant has an SLA of p95 ≤ 3 s. Twenty response times from this morning, sorted, in seconds: 0.7, 0.8, 0.8, 0.9, 0.9, 0.9, 1.0, 1.0, 1.0, 1.1, 1.1, 1.2, 1.2, 1.3, 1.4, 1.5, 1.7, 2.6, 3.8, 4.4. Taking p95 as the 19th of the 20 sorted values, what does the sample say about the SLA?

(A) The p95 is 3.8 s, so this sample breaches the 3 s SLA even though most quotes return in about a second

(B) The p95 is 4.4 s, the slowest of the twenty values, so this sample breaches the 3 s SLA by 1.4 s

(C) The SLA is met: the p95 is 2.6 s, since 18 of the 20 values, or 90% of quotes, are at or below that time

(D) The SLA is met: the mean of the twenty values is about 1.5 s, which sits comfortably inside the 3 s limit

2 WARM-UP

A Sutlej Auto Parts dealer reports that one warranty-claim submission took 9 s at 10:42. The request passed through five services, and each writes its own logs with no field they share, so engineers cannot tie that request's lines together across the services. What should the team add?

(A) A dashboard that shows all five services' log streams side by side on one screen, sorted by timestamp

(B) A latency histogram for each of the five services with p50, p95 and p99, so that the slow service stands out on the dashboard

(C) A standing request that each service team checks its own logs whenever a dealer complains about a slow claim

(D) A trace ID created when a request enters, carried through all five services, with a span per service recording its duration

3 EXAM

Tern Logistics adds a `container_id` label to the stage-latency metric of its Rotterdam yard agent, so that engineers can chart any container's journey. The yard handles about 90,000 containers a month, and the metric already carries labels for 5 pipeline stages and 4 status values. Within two weeks the metrics backend slows to a crawl and its bill triples. What should the team change?

(A) Replace `container_id` with a short hash of it, which shrinks each label value and so reduces what the metrics backend has to store

(B) Sample the metric at 10% of requests, which cuts the data points by about 90% and keeps the backend responsive

(C) Drop `container_id` from the metric labels, keep the bounded ones, and record the container ID in logs and trace attributes

(D) Move to a larger metrics cluster sized for about 1.8 million series, so that per-container charts stay on the dashboard

---

4 EXAM

At Kaveri Freight, Meera's operations dashboard has shown KAVI's mean latency at 310 ms, in green, for weeks, yet dispatchers complain that some replies crawl. Arjun computes percentiles from the same data: p95 is 2,900 ms. He opens the end-to-end trace of one slow ticket: 4,100 ms, most of it inside a single stage. What should change in KAVI's monitoring?

(A) Ask dispatchers to note each slow reply in a shared sheet, and review that sheet with Meera every Friday afternoon

(B) Keep the mean on the dashboard but tighten its alert threshold, so that the panel turns amber as soon as latency starts to drift

(C) Chart p50, p95 and p99 per stage, alert on p95 against the reply SLA, and link each slow request on the dashboard to its trace

(D) Page the on-call engineer for every request slower than the 2,900 ms p95, which means no slow reply goes unseen by the team

---

5 EXAM

Larkspur Health's scheduling assistant handles about 1,200,000 requests a day. Full traces for all of them exceed the storage budget, which allows about 60,000 traces a day, and about 0.4% of requests end in an error. The team must still be able to open the trace of every failed request, and the latency dashboards must stay accurate. Which two changes meet both needs? Select TWO.

[A] Stop storing traces and log each request's total latency instead, since one log line costs far less to store than a trace

[B] Keep 100% of traces but delete each one after 24 hours, which keeps storage small without sampling anything

[C] Compute the latency and error metrics from all 1,200,000 requests before any trace is dropped by sampling

[D] Keep a uniform random 5% of traces, which fits the 60,000 budget exactly and treats every request the same way

[E] Keep the trace of every request that errors, about 4,800 a day, plus a random sample of the rest to fill the budget

---

6 **EXAM** · **AUTOPSY**

Sutlej Auto Parts' warranty-claims pipeline has four stages: a router, retrieval, a drafting model and a checking model. Over a fortnight its Claude bill rose from ₹18,000 to ₹31,000 a day while claim volume held steady, and draft quality must stay at its bar. The only usage metric is total tokens a day for the whole pipeline, so nobody can say which stage grew. What should the team do first?

(A) Add the claim ID and dealer ID as labels on the daily token metric, so that the cost can be traced to individual claims

(B) Record input, output and cache-read tokens and cost for each stage on every request, labelled by stage and model

(C) Ask each stage owner to keep a close eye on their usage and raise anything unusual at the weekly cost review

(D) Cut `max_tokens` by 30% on all four stages at once, which caps spending quickly whichever stage turns out to be responsible

AUTOPSY · the answer is B. Do not choose — dissect. Name the miss family of each wrong option and the clause that kills it.
A _____ **C** _____ **D** _____

7 **EXAM**

Palk Bay Fisheries' export-document agent sends each request through six hops: intake, a router, retrieval, two customs APIs and the model. About one request in twenty misses the 6 s target, at random times and with many requests in flight. Every hop writes good logs, but with no shared field. A senior engineer calls distributed tracing over-engineering for a team of four and proposes matching log lines by timestamp. What should the team do?

(A) Match log lines across the six hops by timestamp: for a small team this gives the same answer as a trace, with no new tooling

(B) Move the agent to a faster model tier, since in most agent pipelines the model is the slowest of the hops in the path

(C) Add tracing: create a trace ID at intake, carry it through all six hops with a span per hop, and keep every slow or failed trace

(D) Add an alert on the 6 s target and ask the on-call engineer to investigate each breach in the hop logs as soon as it happens

8 HEAD-TO-HEAD

Tern Logistics' on-call channel receives about 140 alerts a week from the vessel-ETA service, one for every CPU spike, retry and slow request. Engineers acknowledge 96% of them within a minute and take no action. Last Tuesday a genuine breach of the ETA SLA, p95 ≤ 3 s, went unnoticed for three hours while terminal planners worked from stale arrival times. Which change to the alerting should the team make?

(A) Replace them all with one alert on mean ETA latency above 3 s, which fires far less often and is simple to explain

(B) Ask engineers to read each alert carefully before acknowledging it, and review the week's p95 at Friday's stand-up

(C) Keep a few alerts tied to the SLA, such as end-to-end p95 above 3 s for 10 minutes, each with an owner and a runbook

(D) Keep all 140 alerts and add a second on-call engineer, so that every alert is read by someone with time to act on it

9 HEAD-TO-HEAD

A Larkspur Health clinician reports that her referral at 10:42 took 14 s. For that hour, the team's per-stage dashboards show every stage's p99 inside its target, and the end-to-end p95 is green. Every request carries a trace ID through all stages, and traces are kept for every request slower than 5 s. Where should the engineer look first to find where that referral's time went?

(A) The kept trace for that referral, found by its time and the clinician's session, with each span's duration end to end

(B) A replay of the referral in staging with debug logging switched on, to reproduce the delay under controlled conditions

(C) The per-stage p99 panels for 10:40 to 10:45, zoomed in to find which stage's latency line rose at that moment

(D) A new p99.9 panel for each stage, added to the dashboard so that rarer slow requests like this one become visible

10 HEAD-TO-HEAD

Sutlej Auto Parts' dealer contract promises that 99% of order confirmations each month arrive within 5 s. The team's dashboard alerts when p95 exceeds 5 s, and it stayed green all month with p95 at 3.8 s. Dealers have complained about a handful of very slow confirmations, mostly on large multi-line orders. A month-end query shows p99 at 6.4 s. What should the team conclude and change?

(A) The contract was met: the p95 alert did not fire once all month, and 3.8 s leaves comfortable headroom under the 5 s limit

(B) The contract was breached, so alerts should fire on p50, p90, p95, p99 and p99.9 at once, so that no tail is missed

(C) The contract was breached, so the alert should move to mean confirmation time, which reflects every order, not just the slowest

(D) The contract was breached: it promises a p99, so the alert should fire when p99 exceeds 5 s, the percentile it names

A WORKED SOLUTION · STIMULUS: TRACE

Interrogation IR3-2 · bay 3.4

Try it yourself first: 4 minutes. Then fold and read Holmes solving it aloud.

2 The exhibit shows one hour of the customs-status assistant that Tern Logistics runs for Rotterdam terminal planners. Its SLA is p95 end-to-end ≤ 2.5 s. The operations dashboard charts only the mean and has been green all week, while planners say some answers take 'forever'. Every request carries a trace ID across all four stages. Which reading of the exhibit, and which next step, is right?

```
Tern Logistics · customs-status assistant · Mon 12 Oct, 09:00–10:00
requests 4,600 · dashboard: mean end-to-end 910 ms (green)

stage latency (ms)        p50      p95
router                     35       60
retrieval                 150      240
customs-API call          180    3,900
model                     390      610
end-to-end                760    4,400

slow trace tr-7f3a91: 4,600 ms
  router 38 · retrieval 210 · customs-API 3,742 (2 retries after timeout) · model 610
```

(A) The SLA is breached, so add tracing to all four stages and build a per-stage p95 dashboard before anyone changes a line of code

(B) The SLA is breached, so raise the customs-API retry limit from 2 to 4, so that more timed-out calls succeed in the end

(C) The SLA is met: the mean of 910 ms is well inside 2.5 s, and the 4,600 ms trace is a lone outlier that is not worth chasing

(D) The SLA is breached at a 4,400 ms p95, and the customs-API call holds the tail: fix it first and alert on end-to-end p95

Write your letter and the clause that decided it here, before you turn the page:

_____

✂ HOLMES, ALOUD · READ AFTER YOU HAVE COMMITTED TO A LETTER ✂

The moves

1. Read once: one hour, **4,600** requests, a green mean, a stage table and one slow trace. A green light and angry planners cannot both be telling the whole truth.
2. Underline the tell: 'Its SLA is p95 end-to-end ≤ **2.5 s**.' The promise is about the tail. The dashboard watches the middle.
3. Do the sum: end-to-end p95 is **4,400 ms** against a **2,500 ms** promise, a breach of **1,900 ms**. The **910 ms** mean is true and beside the point.
4. Find the stage: at p50 the model looks largest, **390 ms**. At p95 the customs-API call is **3,900 ms** against **610 ms** for the model. The slow trace agrees: **3,742** of **4,600 ms**, about **81%**, spent in the customs API, including two retries after timeouts.
5. Name the objective: **3.4**. Means hide tails; percentiles per stage and a trace with a shared ID show where one request's time went.
6. Kill the over-build: the trace ID and the per-stage table are already in front of us. Building them again buys nothing.
7. Kill the trade-off reversal: more retries means waiting out more timeouts, which adds time to the tail that is already breaching the SLA.
8. Let the fixed currency pick: latency at the tail. Fix the customs-API call first, and move the alert from the mean to end-to-end p95 so the next breach pages someone.

THE VERDICT

*The dashboard was not lying; it was answering a question nobody had asked. A mean of **910 ms** is quite true and quite useless against a promise about the slowest one in twenty. Read the tail and the case closes itself: **4,400 ms** at p95, and one stage, the customs API, timing out and retrying, owns **3,742 ms** of the slow trace. Fix that call, and teach the alarm to watch the percentile the planners were promised. Then the next breach pages someone before the planners have to complain.*

2. **Answer D**

The tell Its SLA is p95 end-to-end ≤ 2.5 s.

D ✓ **4,400 ms** > **2,500 ms**; at p95 the customs API is **3,900 ms** against **610 ms** for the model, and the slow trace spends **3,742** of **4,600 ms** there.

A OE The trace ID and the per-stage table already exist and already name the stage; more instruments add nothing.

B TR Each retry waits out another timeout, adding time to the very tail that breaches the SLA.

C VG Reads the mean against a p95 promise; the end-to-end p95 is **4,400 ms**, so one request in twenty waits longer than that.

↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

OBJECTIVE 3.5

Bay 3.5 · RAG pipeline: chunking & indexing

Design a RAG pipeline with appropriate chunking and indexing strategies

What this bay drills. Cut along headings, keep a table with its header, add a little overlap, prepend each chunk's context before embedding and BM25 indexing, re-index what changed. Under ~200k tokens, skip RAG and cache.
Law 2: *Chunk along meaning, not length.*
10 items · 3 warm-up · 4 exam · 3 head-to-head · about 22 minutes · bar ≥ 8 / 10

1 WARM-UP

Chilika Cold Stores tests retrieval over its handling manuals. With plain embeddings, the right chunk is missing from the top 20 in 5.7% of test questions. The team plans to add contextual embeddings, contextual BM25 and a reranker, following Anthropic's published contextual-retrieval results. What top-20 failure rate should they plan around?

(A) About 1.9%, since each stage takes a further bite and all three together cut top-20 retrieval failures by roughly two-thirds

(B) About 0.6%, since the published cuts of 35%, 49% and 67% apply one after another, each to what the last stage left

(C) About 2.9%: a cut of roughly half, since a reranker only reorders the final 20 and cannot change which chunks reach the model

(D) About 3.7%, a cut of roughly one-third, since contextual embeddings carry the gain and contextual BM25 mostly repeats the same match

2 WARM-UP

Hampi Heritage Hotels wants its concierge assistant to answer from the house guide: 38 documents, about 90,000 tokens in total, revised twice a year. The developer has drafted a plan for chunking, embeddings, a vector index and a reranker. What is the simplest design that answers well?

(A) Build the planned pipeline with heading-aware chunks and a reranker; this keeps the design ready for the day the guide grows

(B) Put the whole guide in the prompt as a stable prefix and cache it; at this size the corpus does not need a retrieval pipeline

(C) Send the whole guide in the prompt on every call but leave caching off: twice-yearly revisions would keep invalidating any cache

(D) Embed each of the 38 documents whole as one vector in the index, so that the team avoids making any chunking decisions

3 WARM-UP

Early in the project, Kaveri cut its 400 heading-structured SOPs into fixed-size chunks. For KF-2481 the retriever returned the cancellation steps instead of SOP 214 §4, customs e-way bill re-filing, because the re-filing procedure had been cut across two chunks. Which chunking change fixes this?

(A) Chunk along the SOP headings so that §4 stays one unit, with a small overlap carrying context across each boundary

(B) Add an instruction telling KAVI to read the neighbouring chunks whenever a retrieved procedure looks incomplete

(C) Keep the fixed-size chunks and rebuild the vector index with a stronger embedding model that separates cancellation from re-filing text

(D) Double the fixed chunk size so that the whole of §4 fits inside one chunk along with the sections either side of it

---

4 EXAM

At Kaveri, Arjun has moved the 400 SOPs to heading-aware chunks, so SOP 214 §4 (customs e-way bill re-filing) is now one clean chunk. Yet for KF-2481 the retriever still prefers similar-looking re-filing sections from other SOPs. Reading the chunks, Arjun sees that the chunk text never says which SOP or which procedure it belongs to; it opens at its first numbered step. Dispatchers describe the situation, not SOP numbers. What should Arjun add?

(A) Contextual retrieval: prepend a short context naming the SOP, its section and the procedure to each chunk before embedding and BM25 indexing

(B) A metadata filter on SOP number: a question tagged with SOP 214 can then return only chunks drawn from SOP 214's own sections

(C) A firm rule in KAVI's system prompt to confirm which SOP each retrieved paragraph comes from before quoting its steps to a dispatcher

(D) A larger embedding model and a rebuilt vector index, since stronger vectors separate similar-looking re-filing sections from different SOPs

---

5 **EXAM** · **AUTOPSY**

Chilika Cold Stores' picking assistant answers from 540 client handling sheets held in a vector index; about 25 sheets are revised each week. The index was built once at launch and revised sheets are simply appended. Last week a picker asked for the longest time a frozen-dessert pallet may stay outside the chamber and got the March figure, though the sheet had been revised four days earlier; both versions' chunks sit in the index and score almost equally. What is the best fix for the indexing pipeline?

(A) Add a line to the system prompt telling the assistant to prefer the more recent figure whenever two retrieved chunks disagree about a limit

(B) Re-index the changed sheets every week, label each chunk with its sheet version, and remove superseded chunks when the new version lands

(C) Put a reranker behind the vector search, so that the current version's chunk is scored above the older one before Claude ever sees it

(D) Rebuild the entire index from scratch once a quarter, which clears out every old version in a single clean pass and keeps the pipeline simple

AUTOPSY · the answer is B. Do not choose — dissect. Name the miss family of each wrong option and the clause that kills it.
A _____ **C** _____ **D** _____

6 **EXAM**

Shivalik Pharma indexes 1,200 stability-testing SOPs in fixed 400-token chunks. Each SOP carries storage-condition tables of up to 40 rows, with columns for temperature, humidity, pull point and acceptance limit. Analysts asking for the pull points at 30 °C / 65% RH for a product line get answers that quote the right row but misread its columns, because the chunk holding that row does not hold the table's header row. Which change fixes the retrieval?

(A) Rebuild the vector index with a larger embedding model, which represents numeric rows like 30 °C / 65% RH more precisely than the current one

(B) Add a 50-token overlap between the fixed chunks: it carries the end of each chunk into the next, so no row is cut in half

(C) Chunk tables as tables: keep each table with its header row in one chunk, repeating the header in every piece when a table must be split

(D) Raise top-k from 10 to 40, so that the chunk carrying the header row is likely to be retrieved alongside the chunk carrying the matching row

7 EXAM

Hampi Heritage Hotels is indexing 260 front-desk SOPs. They are heading-structured, and many paragraphs switch between Kannada and English mid-sentence; staff ask questions in Kannada, English or a mix. A vendor proposes splitting every paragraph at each language switch, keeping Kannada and English chunks in two separate indexes, and routing each question by its detected language. Which two choices should the architect make instead? Select TWO.

- **A** Use keyword-only BM25 search, since exact term matching avoids the confusion that mixed-language text can cause for embedding models
- **B** Build one shared index with an embedding model that handles both languages, so a Kannada question can match a step written mostly in English
- **C** Use fixed 300-token chunks with a 20% overlap, which keeps chunk sizes even across both languages and carries each language switch across the boundary
- **D** Chunk along the SOP headings and keep each mixed-language paragraph whole, since a step that changes language mid-sentence is one unit of meaning
- **E** Translate every SOP into English before chunking and discard the original text, so that the index holds one clean language that embeds consistently

8 HEAD-TO-HEAD

Shivalik Pharma's regulatory-affairs team wants Claude to answer from its dossier library: 2,100 documents, about 640,000 tokens, revised monthly. The model in use has a 1M-token context window, and an engineer proposes loading the whole library into a cached prompt prefix instead of building retrieval, arguing that it fits and rarely changes. Analysts ask about 3,000 questions a day. Which design should the architect choose?

- (A) Build retrieval with heading-aware chunks, hybrid search and reranking: a library this size is far past the point where whole-prompt loading pays
- (B) Build retrieval and also load the whole library as a cached prefix, so that the model can check every retrieved chunk against the full source
- (C) Split the library into four cached prefixes of under 200,000 tokens each and send all four on every call to stay inside the limit
- (D) Load the whole library as a cached prefix: it fits inside the 1M-token window, and monthly revisions leave the prefix unchanged between updates

9 HEAD-TO-HEAD

Ostrava Rail indexes 18,000 track-inspection reports. Inspectors write them as continuous free text of 2,000–6,000 tokens, with no headings, numbered steps or section breaks. The pipeline uses fixed 500-token chunks, and an observation that straddles a chunk boundary is retrieved in halves: the defect in one chunk, the measurement and the action taken in the next. Which chunking change helps most?

(A) Raise the chunk size to 2,000 tokens so that most observations fall inside one chunk instead of straddling a boundary

(B) Keep fixed-size chunks but add an overlap between neighbours, so that an observation cut at one boundary still appears whole in the next chunk

(C) Switch to heading-aware chunking, which follows the document's own structure and is the usual recommendation for long technical documents

(D) Add contextual retrieval, prepending a short report-level summary to every chunk before embedding so that each fragment knows its source report

---

10 HEAD-TO-HEAD

Chilika Cold Stores' retrieval already uses contextual embeddings with contextual BM25, and its top-20 failure rate on the test set is 2.9%. Logs show that when an answer is wrong, the right chunk was usually retrieved somewhere in the top 150 but ranked below 20. Each answer may carry at most 20 chunks to keep the per-call bill flat. What should be added?

(A) Ask Claude in the system prompt to say when none of the 20 chunks answers the question, and log those cases for a weekly review

(B) Contextual retrieval, prepending a chunk-specific context to every chunk before both the embedding and the BM25 indexing run

(C) Raise top-k from 20 to 150: the chunk currently ranked below 20 is then passed to Claude along with the rest of the candidates

(D) A reranker: retrieve about 150 candidates, rescore each one against the question, and pass only the top 20 to Claude

---

A WORKED SOLUTION · STIMULUS: ARCHITECTURE

Interrogation IR3-3 · bay 3.5

Try it yourself first: 4 minutes. Then fold and read Holmes solving it aloud.

3 A Chilika Cold Stores supervisor asked for the longest door-open time for ice cream in Chamber 6 and was given the humidity figure from the same row. The exhibit describes the chamber-SOP pipeline and its week-41 failure log. The platform team can ship one change before the Diwali stock build-up. Which single change fixes the failures in that log?

```
CHILIKA COLD STORES · chamber-SOP assistant · pipeline v1
Corpus    620 SOPs, H1–H3 headings; ~250 carry set-point tables
          (1 header row + up to 30 rows: product, °C, RH%, door-open min)
Updates   ~30 SOPs revised each Friday; full re-index on the 1st of the month
Chunking  fixed 500 tokens, no overlap
Index     contextual embeddings + contextual BM25, scores fused
Retrieve  top 150 → reranker → top 20 to the model

Week-41 failure log: 23 wrong answers, all on table values
  21 / 23  retrieved chunk held the matching rows but not the header row;
           the model read the columns in the wrong order
   2 / 23  header present, but the matching row was split across two chunks
  23 / 23  answer drawn from an SOP not revised since the last re-index
  reranker placed the matching rows in the top 5 every time
```

(A) Rebuild the vector index with a larger embedding model that represents numeric table rows, such as °C and RH% values, more faithfully

(B) Replace the monthly full re-index with a Friday re-index of the changed SOPs, version-labelling each chunk and dropping every superseded version

(C) Widen the rerank pool from 150 to 300 and pass the top 40 to the model, so that the chunk holding each table's header arrives alongside its rows

(D) Re-chunk along the SOP headings, keeping each table with its header row and repeating the header in any piece of a split table, then re-index

Write your letter and the clause that decided it here, before you turn the page:

_____

✂ HOLMES, ALOUD · READ AFTER YOU HAVE COMMITTED TO A LETTER ✂

Flag and move. On the first pass Holmes could not separate two options in under a minute, so he marked his best letter, flagged it and moved on. Watch the moves for where he returns and what settles it.

The moves

1. Read once: heading-structured SOPs with **30**-row tables, Friday revisions against a monthly re-index, fixed **500**-token chunks, and the full hybrid + rerank stack already in place. Asked: the ONE change that fixes the week-41 log.
2. Name the objective: **3.5**, chunking and indexing. Retrieval is not the suspect; ~**150** → rerank → top **20** is already the canon stack.
3. Kill the two easy ones: the **300**-pool / top-40 option spends tokens and latency on a gamble (TR), and the bigger embedding model mends the index, not the chunk (VG).
4. Flag and move: two survivors, and both describe true faults. Fixed **500**-token cuts across **30**-row tables break law two; a monthly re-index under Friday revisions breaks the weekly rule. I cannot separate them on a first read, so I pencil in the re-chunk, flag the item and move on.
5. Return with fresh eyes and read the log, not the design. Underline the tell: **21** of **23** — 'retrieved chunk held the matching rows but not the header row'.
6. Settle it with the second log line: **23** of **23** answers came from 'an SOP not revised since the last re-index'. Staleness caused none of them. The Friday re-index is a real defect, for the next ticket.
7. Let the failure pick: chunk along the headings, keep each table with its header row (repeating it in any split piece), re-index. Unflag and confirm.

THE VERDICT

The design had two faults and the paper offered me both. A fault is not a cause. The log said the reranker had found the right rows and the model could not read them, because the chunker had cut the header away; it also said every bad answer came from an SOP the last re-index had already caught. So the chunker did it. Chunk along the headings, keep each table with its header, re-index — and write the Friday re-index on the next ticket. Flagging cost me ninety seconds; choosing between two true statements on a first read would have cost me the mark.

3 **Answer D**

The tell retrieved chunk held the matching rows but not the header row

D ✓ All **23** failures trace to fixed cuts through tables: **21** lost the header, **2** split a row. Table-aware, heading-aware chunks fix both.

A **VG** Index vs chunk: the header is missing from the chunk text, and no embedding can put it back.

B **CM** A real defect, but not this log's: all **23** wrong answers came from SOPs not revised since the last re-index.

C **TR** Doubles context and latency on a gamble; the rows were already in the top **5,** and nothing pairs a header with its rows.

↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A2 ; Dock Walk Station 1 ; Dock Gym station 3.5

OBJECTIVE 3.6

Bay 3.6 · Retrieval strategy

Apply retrieval strategies matched to data shape and query pattern

What this bay drills. An ID goes to a structured lookup, a count to an aggregate, a policy question to hybrid retrieval. Exact codes lean keyword; paraphrases lean semantic. Rerank the shortlist.
Law 3: *Retrieval follows the shape of the data, not the fashion.*
10 items · 3 warm-up · 4 exam · 3 head-to-head · about 22 minutes · bar ≥ 8 / 10

1 WARM-UP

Ostrava Rail's depot assistant sends every question through one hybrid RAG pipeline. Fitters most often ask things like 'what is the current status of work order WO-58213?', and the answers sometimes quote an older status from a weekly report that happens to mention the order. Where should questions of this kind go?

(A) To a new agent that searches the reports, the work-order system and the fitters' notes, then reconciles the three answers

(B) To a structured lookup that queries the work-order system by its ID and returns the live record, with no retrieval step at all

(C) To the hybrid pipeline as now, with the weekly reports re-indexed every night so that any status they quote is at most a day old

(D) To the keyword side of the hybrid pipeline only, since BM25 matches an exact work-order ID far better than the embedding side does

---

2 WARM-UP

Chilika Cold Stores audits 50 wrong answers from its assistant. In 46 of them the chunk holding the right answer was among the 20 passed to Claude, but on average only 2 of those 20 chunks were about the question at all; the rest were look-alike procedures from other chambers. Which retrieval property is weak?

(A) Recall, so top-k should rise from 20 to 50 to make sure that the right chunk is reliably somewhere in the set passed to Claude

(B) Recall: the right chunk must be ranked too low, since 18 of every 20 retrieved chunks are about other chambers' procedures

(C) Precision: about 2 relevant chunks in 20, or 10%, buries the right chunk among look-alikes, while recall holds at 46 of 50

(D) Neither: the right chunk was present in 46 of 50 cases, which leaves the model's reasoning as the only remaining cause

---

3 WARM-UP

Chilika Cold Stores' maintenance assistant searches its equipment manuals with BM25 keyword matching only. Technicians describe what they see, such as 'the door won't close properly and ice keeps forming on the rubber edge', while the manuals say 'door gasket frost build-up'. Most such questions return nothing useful. Which retrieval change fits this query pattern?

(A) Add a metadata filter on equipment model to the BM25 search, so that each question is searched only against the manual for the unit under repair

(B) Add embedding search beside BM25 and fuse the two score lists, so a description can match the manual's wording by meaning

(C) Give technicians a printed list of the manuals' exact fault names and ask them to search with those terms from now on

(D) Re-chunk the manuals along their headings with a small overlap, so that each fault section is retrieved as one complete unit

---

4 EXAM

Hampi Heritage Hotels' finance team asks its assistant questions such as 'what was the total value of complimentary room upgrades at each of our 14 properties in September?'. The assistant answers from hybrid RAG over guest folios and monthly reports, passing 20 chunks per question, though September alone holds about 4,200 upgrade lines; its totals are fluent, confident and different every time the question is asked. The folios also sit, line by line, in the property-management database. What is the best change?

(A) Route totals and counts to a SQL-style aggregate over the folio table in the database, and keep RAG for policy and procedure questions

(B) Raise top-k from 20 to 200 so that far more folio lines reach the model, giving it enough rows to add up a total for each property

(C) Ask the model to show its working and state its confidence for each total, and have finance spot-check the figures every month

(D) Re-chunk the folios so that each chunk holds exactly one folio line, which lets retrieval return precise rows instead of mixed passages

---

5 **EXAM** · **AUTOPSY**

Shivalik Pharma's index holds every version of its 1,200 SOPs, including drafts under review and versions that have been withdrawn; each chunk carries a status label (draft, effective, withdrawn) and an effective date. An operator asked for the line-clearance steps on Line 4 and was given steps from a draft that had never been approved, which the assistant presented as current. Retrieval ranked the draft first because its wording matched the question best. What is the best fix?

(A) Apply a metadata filter at query time so that only chunks labelled effective, and in force today, can be retrieved for operators

(B) Instruct the assistant to warn operators whenever a retrieved procedure might be a draft, and to suggest they confirm it with QA

(C) Build a separate index for each status and send every question to all three, letting the model compare the versions it gets back

(D) Add a reranker over the top 150 results, so that the current approved procedure, not a draft, rises to the top before the 20 reach the model

AUTOPSY · the answer is A. Do not choose — dissect. Name the miss family of each wrong option and the clause that kills it.
B _____ **C** _____ **D** _____

6 **EXAM**

Ostrava Rail's signalling assistant answers from the rule book and 300 safety notices using embedding search alone; each rule is already its own chunk. Controllers ask in two ways: by exact reference ('what does Rule 47.3.2 require?') and by description ('what do I do when a signal shows no light at all?'). Descriptions are answered well; exact references often return the neighbouring rule, 47.3.1 or 47.4.2. Which change fits both query patterns?

(A) Swap in a larger embedding model trained on regulatory text, which separates Rule 47.3.1 from Rule 47.3.2 better than a general model

(B) Keyword search: replace the embeddings with BM25, which matches exact references such as Rule 47.3.2 far more reliably than vectors do

(C) Send reference questions to one agent and description questions to another, each with its own index, tools and system prompt

(D) Hybrid search: add BM25 beside the embeddings, fuse the two score lists, and rerank the fused shortlist before passing the top 20

7 EXAM

Kaveri's dispatcher assistant gets three kinds of question: 'where is KF-2481 right now?', 'how many customs holds did we clear this week?' and 'what do I do when an e-way bill expires in transit?'. The route SLAs are structured lookup ≤ 2 s, policy (RAG) question ≤ 4 s and aggregate ≤ 5 s. Today all three go through hybrid RAG over the 400 SOPs and the shipment notes. Which two routing decisions are right? Select TWO.

- **A** Send 'where is KF-2481 right now?' to a structured lookup on the shipment record by consignment ID, within the 2 s SLA
- **B** Send 'how many customs holds did we clear this week?' to an SQL-style aggregate over the exceptions table, within the 5 s SLA
- **C** Keep the customs-holds count in hybrid RAG but raise top-k to 150, so that every hold cleared this week reaches the model to be counted
- **D** Send the KF-2481 question through keyword-only BM25 over the shipment notes, since an exact consignment ID is a keyword-shaped query
- **E** Take the e-way bill question out of RAG and send it to a structured lookup as well, since that route answers inside 2 s

---

8 HEAD-TO-HEAD

A front-desk manager at Hampi Heritage Hotels asks the assistant: 'Can booking HH-30917 be refunded under our monsoon cancellation policy?' The booking lives in the reservation system; the policy is a four-page document with exceptions for heritage-festival dates, held in the policy library behind hybrid RAG. Today the question goes only to RAG, and the answer quotes the policy correctly but guesses the booking's dates. How should this question be served?

- **A** Fetch booking HH-30917 from the reservation system by its ID, retrieve the monsoon policy through hybrid RAG, and answer from both
- **B** Send it to the structured lookup only, since it names a booking ID, and return the booking's dates and its refund status
- **C** Hand the question to an autonomous agent with access to every hotel system, letting it decide which records and documents to read
- **D** Index the booking records alongside the policy in hybrid RAG, so that BM25 matches HH-30917 exactly and both parts come back together

---

9 HEAD-TO-HEAD

At Kaveri, dispatchers type consignment IDs such as KF-2481 into the assistant to get the latest status. Arjun's log shows these consignment-ID lookups going through RAG instead of a structured lookup: they cost ₹1,800 a day against ₹4 for the right drawer, and the structured-lookup SLA is ≤ 2 s. A colleague suggests raising the BM25 weight in the hybrid fusion so that IDs match exactly. What should Arjun do?

(A) Cache each consignment's RAG answer for 24 hours, which cuts the ₹1,800 a day and answers repeated lookups well inside 2 s

(B) Add a reranker after the hybrid search, which lifts the shipment note holding the exact consignment ID to the top of the 20

(C) Raise the BM25 weight in the hybrid score fusion so that an exact consignment ID reliably outranks semantically similar shipment notes

(D) Detect consignment IDs before any retrieval runs and send those questions straight to a structured lookup on the shipment record

10 HEAD-TO-HEAD

Ostrava Rail's depot assistant answers from local working instructions written separately by each of its 9 depots; the same task often has slightly different steps at each depot. Fitters rarely name their depot in a question, but the app always knows which depot a fitter is signed in at. Fitters at Depot 3 keep getting Depot 7's steps for wheel-lathe set-up. Which change fixes this most reliably?

(A) Raise top-k so that the steps from all 9 depots reach the model, and let it pick the version that matches the fitter's depot

(B) Add a line to the system prompt asking the assistant to check which depot an instruction comes from before giving any steps

(C) Add contextual retrieval, prepending the depot name and task to each chunk before embedding, so every chunk carries the depot it belongs to

(D) Tag every chunk with its depot and apply a metadata filter on the signed-in fitter's depot, so only that depot's steps can be retrieved

OBJECTIVE 3.7

Bay 3.7 · Connection protocols

Evaluate connection protocols; select the integration mechanism

What this bay drills. One fixed call from your own code: API or CLI. Many tools, many clients, reuse across apps: an MCP server (tools, resources, prompts; stdio or Streamable HTTP). A partner with its own goals: agent-to-agent.
Law 1: *Every tool you add is a door; know who holds the key.*
7 items · 2 warm-up · 3 exam · 2 head-to-head · about 16 minutes · bar ≥ 6 / 7

1 WARM-UP

Every night at 02:00 a scheduled script at Chilika Cold Stores makes one call to the electricity board's tariff API, fetches the next day's single tariff file and loads it into the energy-planning database. No model reads or decides anything in this job, the endpoint has not changed in three years, and no other application uses the data. Which integration mechanism fits?

(A) Wrap the tariff API in an MCP server, so that the script and any future Claude application reach it through one standard connector

(B) Set up agent-to-agent messaging with the electricity board's system, so that it can push the tariff file as soon as it is ready

(C) Call the tariff API directly from the script with an ordinary HTTP client, keeping the one fixed request in the team's own code

(D) Give a Claude agent the tariff API as a tool each night; the agent can then notice and adapt if the file format ever changes

2 WARM-UP

Shivalik Pharma's developers add an MCP server named `lims` to the checked-in `.mcp.json`. It exposes three tools: `get_batch_record`, which is read-only, and two that change data. The team wants Claude Code to run `get_batch_record` without asking, for every developer, and to keep asking before the other two. Which permission rule should go in the shared project settings?

(A) An allow rule for `mcp__lims__get_batch_record`: the mcp prefix, the server name and the tool name, joined by double underscores

(B) An allow rule for `get_batch_record` on its own, since the tool name is already unique across every server the team has configured

(C) An allow rule for `lims__get_batch_record`, the server name and tool name joined, as they appear together in the `.mcp.json` file

(D) An allow rule for `mcp__lims`: it covers every tool the server exposes and saves updating the rule whenever a tool is added

3 **EXAM**

Hampi Heritage Hotels runs four Claude-based applications: a Claude Desktop pilot for front-desk managers, Claude Code for its 9 developers, an Agent SDK service that re-books guests after cancellations, and a guest chatbot. All four need the same 6 property tools: room status, rates, folio lookup, housekeeping, transfers and guest preferences. Each application has its own hand-written integration for each tool, so 24 integrations must be patched whenever the property-management API changes, as it did twice last quarter. What should the architect do?

(A) Build one MCP server that exposes the 6 tools and connect all four applications to it as clients, so an API change is fixed in one place

(B) Connect each application agent-to-agent with the property-management vendor's system, letting that system decide how to answer each request

(C) Replace the four applications with one agent that holds all 6 tools and serves managers, developers, re-booking and guests together

(D) Move the 24 integrations into one shared code library that every application imports, then redeploy all four after each API change

---

4 **EXAM**

Arjun is finishing `kaveri_mcp.py`, the FastMCP server that both KAVI and the dispatchers' desktop client will connect to. KAVI decides for itself when to check a consignment's status or look for another carrier. The desktop client attaches the text of SOP 214 when a dispatcher opens a customs case, and dispatchers pick the customs triage checklist themselves from a menu, entering the consignment ID. Which two primitive choices are right? Select TWO.

[A] Write the customs triage checklist into KAVI's system prompt, so that the agent applies it to every case without dispatchers needing a menu

[B] Expose SOP 214 as a tool, `read_sop_214`, which KAVI calls on its own whenever it judges that a case involves an e-way bill

[C] Expose `customs_triage` as a prompt, a template the dispatcher picks from the menu and fills in with the consignment ID

[D] Expose `get_shipment_status` and `find_alternate_carriers` as tools, since KAVI decides when to call them and with which arguments

[E] Split the server in two, one holding KAVI's tools and one holding the desktop client's SOPs and checklists, each deployed separately

---

5 EXAM

Ostrava Rail is building a remote MCP server for its 30 depot applications against the current MCP specification (2026-07-28). The contractor's design follows a 2024 tutorial: the HTTP+SSE transport, and a session that the server creates when a client connects and uses to remember that client's context between requests. Depot applications will call the server over the company network. What should the architect change in the design?

(A) Switch the transport to stdio, which replaced HTTP+SSE in the current specification and removes the need for server-side sessions altogether

(B) Keep HTTP+SSE, which the tutorial shows working end to end, and add OAuth so that each of the 30 depot applications is properly authenticated

(C) Use Streamable HTTP, build the server stateless with every request carrying its own `_meta`, and authenticate each depot application with OAuth

(D) Keep the server-side sessions but replicate the session store across three nodes, so that no depot loses its context when a node restarts

6 HEAD-TO-HEAD

Shivalik Pharma has built an MCP server that reads chromatography results from the lab's instrument database. The pilot ran on one analyst's workstation and connected over stdio. It is now to serve 40 analysts at three sites through one copy running in the data centre, and QA wants every analyst's access checked and recorded by the server. Which connection design fits?

(A) Serve the one copy over Streamable HTTP, with OAuth so each analyst's client authenticates and the server can check and log access

(B) Serve it over Streamable HTTP behind one long-lived shared API key that the lab IT team issues once to every analyst's client

(C) Serve it over HTTP with server-sent events, the remote MCP transport built for streaming results from a shared server to many clients

(D) Keep stdio and install a copy on each of the 40 workstations, so that every instance inherits that analyst's own login and permissions

7 HEAD-TO-HEAD

Ostrava Rail shares lines with a partner freight operator. When an engineering possession closes a section, Ostrava's planning agent must agree new freight paths with the partner. The partner runs its own planning agent, which owns its fleet rules and tools and decides by its own goals which paths it will give up; it publishes no booking endpoints for outsiders. Which integration mechanism fits?

(A) MCP: Ostrava builds a server around the partner's planning functions and exposes them as tools for its own planning agent to call

(B) Agent-to-agent: Ostrava's planning agent negotiates with the partner's agent, which applies its own rules and calls its own tools

(C) A joint planning agent run by both companies that holds both fleets' tools and rules and replans every shared freight path centrally

(D) Direct API calls from Ostrava's planning code into the partner's booking system, with fixed rules deciding which freight paths to request

OBJECTIVE 3.8

Bay 3.8 · Progressive vs monolithic context

Evaluate progressive discovery vs. monolithic context strategy

What this bay drills. Large, changing, rarely used: send the catalogue, fetch on demand (tool search with defer_loading). Small, stable, always used: send it whole and cache it. Below the cache minimum nothing is cached.
Law 5: *Send the map first, the territory on demand.*
7 items · 2 warm-up · 3 exam · 2 head-to-head · about 16 minutes · bar ≥ 6 / 7

1 WARM-UP

Arjun builds KAVI's progressive-discovery context. Each call sends 9 tool stubs of 25 tokens each, one 260-token header and 20 SOP summaries of 180 tokens each, then fetches one 120-token SOP section on demand. The monolithic alternative, every schema and every SOP on every call, is 362,460 tokens. What does a progressive call carry, and how large is the gap?

(A) 1,105 tokens, about 328 times smaller, since each SOP summary costs the same 25 tokens as one tool stub

(B) 4,085 tokens: about 89 times smaller, since the on-demand fetch is not part of the context that each call carries

(C) 4,205 tokens: 225 for the stubs, 260 for the header, 3,600 for the summaries and 120 for the fetch, about 86 times smaller

(D) 6,285 tokens, about 58 times smaller, since each of the 9 tool stubs is sent with its own 260-token header

2 WARM-UP

Ostrava Rail's ticket-classification step runs on Haiku 4.5, about one call every 5 seconds. A 3,000-token block of tool schemas and system prompt, identical on every call, carries a cache breakpoint, yet the usage fields show zero cache-creation and zero cache-read tokens on every call. What explains this?

(A) The 3,000-token prefix is below the 4,096-token minimum cacheable length for Haiku 4.5, so nothing is cached and no error is raised

(B) Caching covers only the messages array, so a breakpoint on the tools and system prompt is ignored and must move to the last user turn

(C) The 5-minute cache lifetime runs out between calls, so every request has to write the prefix again before a later one can read it

(D) The smallest model tier does not support prompt caching at all, so the step must move to a mid-tier model before anything can be cached

3 **EXAM**

Arjun measures two context strategies for KAVI. Monolithic, all nine tool schemas and all 400 SOPs on every call, is 362,460 tokens; progressive discovery is 4,205 tokens a call. Devika holds a ₹6 ceiling per resolved exception, and Rajan asks whether prompt caching, with cache reads billed at 0.1× of the input price, would make the monolithic design competitive. The SOPs change weekly. What should Arjun reply?

(A) Yes: a cached read bills about 36,246 token-equivalents a call, close enough to the progressive 4,205 to keep the simpler design

(B) Yes: once cached, the 362,460-token prefix is free on every later call, so the two strategies cost almost the same per exception

(C) No: a cached read still bills about 36,246 token-equivalents a call, roughly 8.6 times the progressive call, so progressive discovery stays

(D) Neither: split the 400 SOPs across four subagents of 100 each, so every subagent's cached monolithic context stays under 200,000 tokens

---

4 **EXAM**

Ostrava Rail's network-operations agent connects to 12 MCP servers exposing 140 tools; an incident typically uses 4 or 5 of them, and the servers add or change tools most weeks. Incidents run for hours, and each log-search result is 8,000–15,000 tokens that is no longer needed once read, so input passes 100,000 tokens by mid-incident. The cost per incident must fall. Which two changes fit? Select TWO.

[A] Enable `clear_tool_uses_20250919`, which clears old tool results once input passes about 100,000 tokens and keeps the 3 most recent

[B] Give each of the 12 MCP servers its own subagent, with an orchestrator choosing which subagent handles each step of the incident

[C] Send all 140 schemas whole on every call behind a cache breakpoint, since cached reads cost a tenth of the input price or less

[D] Enable `clear_thinking_20251015`, so that earlier reasoning blocks are cleared from the context once the incident passes 100,000 tokens

[E] Mark the rarely used tools `defer_loading: true` and add the tool search tool, so their schemas load only when a search finds them

---

5 **EXAM**

Hampi Heritage Hotels' guest-service app builds its tools array per request: a router picks the 6–10 of 45 tools it judges relevant, in whatever order they come back, and a 2,000-token system prompt follows with a cache breakpoint. Requests arrive every few seconds; cache-creation tokens appear on almost every request, yet cache-read tokens stay close to zero. What should the architect change?

(A) Send one fixed tools array in a fixed order on every request, with rarely used tools marked `defer_loading: true` for the tool search tool

(B) Switch to the 1-hour cache lifetime, which lets cached prefixes survive the quiet gaps between guest requests through the night

(C) Move the cache breakpoint onto the last user message, so that the cached prefix covers the conversation as well as the tools and system prompt

(D) Keep the router but sort its chosen tools alphabetically, so that requests picking the same tools produce an identical prefix and share a cache entry

---

6 HEAD-TO-HEAD

Chilika Cold Stores' warehouse agent has 12 tools whose schemas average 800 tokens, 9,600 in all; they change about once a quarter, and an average call uses 9 of the 12. A consultant proposes progressive discovery: a 500-token catalogue of one-line stubs, with each schema fetched on first use in the call. Calls arrive every few seconds, and cached reads bill at 0.1× of input. Which design costs less per call?

(A) Use the tool search tool with `defer_loading: true` on all 12 tools, the API-native catalogue that saves writing custom fetch code

(B) Send all 12 schemas whole but leave caching off, since a cached prefix is re-written at 1.25× the input price on every call

(C) Adopt the catalogue, which cuts the tool prefix from 9,600 tokens to 500 and so saves about 9,100 tokens on every single call

(D) Keep all 12 schemas whole as a cached prefix: about 960 token-equivalents a call, against about 7,700 for the catalogue plus nine fetches

---

7 HEAD-TO-HEAD

Shivalik Pharma's batch-review agent calls `get_batch_record` once at the start of a session; its 6,000-token result holds the batch's reference limits, needed until the session ends. It then runs 30 or more `search_deviation_log` calls of about 10,000 tokens each. The team enabled `clear_tool_uses_20250919` with defaults: clearing starts near 100,000 input tokens and keeps the 3 most recent tool uses. Late in sessions the agent now misquotes the limits. What should change?

(A) Keep the clearing on but list `get_batch_record` in `exclude_tools`, so its result is never cleared while old log searches still are

(B) Also enable `clear_thinking_20251015`, so that earlier reasoning blocks are cleared and the room freed lets the batch record stay

(C) Tell the agent in its system prompt to restate the batch's limits in every reply, so that they stay in context after the result is cleared

(D) Raise the number of tool uses kept from 3 to 10, so that the batch record is more likely to survive until the end of each session

---

A WORKED SOLUTION · STIMULUS: NUMBERS

Interrogation IR3-4 · bay 3.8

Try it yourself first: 4 minutes. Then fold and read Holmes solving it aloud.

4 Shivalik Pharma runs one QA agent at two desks and must pick a context strategy for each: the monolithic block sent whole behind a cache breakpoint, or progressive discovery. The day desk is busy all shift; the night desk is quiet, and its 8 calls a night typically arrive 70–90 minutes apart. Using the exhibit, which plan gives the lowest daily input bill?

```
SHIVALIK PHARMA · QA agent · context options (input tokens per call)
Monolithic   24 tool schemas + 6 reference SOPs = 18,000 tokens, changes quarterly
Progressive  catalogue 1,400 + on average 3 fetches × 900 = 4,100 tokens, sent uncached
Cache        5-minute TTL; write 1.25×, read 0.1× of the input price

Desk                   Calls/day   Typical gap between calls
Day batch release      5,200       a few seconds
Night deviation desk   8           70–90 minutes
```

(A) Progressive discovery at the day desk, where 5,200 calls a day make every token count, and monolithic with caching at the night desk, where 8 calls cost little

(B) Monolithic with caching at the day desk, about 1,800 token-equivalents a call against 4,100, and progressive discovery at the night desk, 4,100 against 22,500

(C) Progressive discovery at both desks: its 4,100 tokens a call is under a quarter of the 18,000-token block that monolithic sends

(D) Monolithic with caching at both desks, since a cached read bills about 1,800 token-equivalents a call, which is under half of progressive discovery's 4,100

Write your letter and the clause that decided it here, before you turn the page:

_____

✂ HOLMES, ALOUD · READ AFTER YOU HAVE COMMITTED TO A LETTER ✂

The moves

1. Read once: one agent, two desks, two strategies. Asked: the cheaper strategy per desk, measured as a daily input bill.
2. Name the objective: **3.8**, progressive discovery vs monolithic + caching. The answer is a crossover, not a slogan.
3. Price the monolithic call both ways: warm read **18,000 × 0.1 = 1,800**; cold write **18,000 × 1.25 = 22,500**. Progressive is **4,100** either way, since it is sent uncached.
4. Find the crossover: h × **1,800** + (**1** − h) × **22,500** = **4,100** gives a cache hit rate h ≈ **0.89**. Monolithic wins only where the cache is warm on about **9** calls in **10**.
5. Underline the tell and place each desk: the day desk (**5,200** calls, seconds apart) sits at h ≈ **1,** so monolithic; the night desk ('its **8** calls a night typically arrive **70–90** minutes apart') against a **5**-minute TTL sits at h ≈ **0,** so progressive.
6. Kill the slogans with daily totals: monolithic everywhere = **9,360,000** + **180,000** = **9,540,000**; progressive everywhere = **21,320,000** + **32,800** = **21,352,800**; the reversed plan = **21,320,000** + **180,000** = **21,500,000**; the split plan = **9,360,000** + **32,800** = **9,392,800**.
7. Check the escape hatch before signing: a **1**-hour cache would not rescue the night desk, because **70–90**-minute gaps outlast it too and each write would bill **2×** = **36,000**.

THE VERDICT

*The table did not ask which strategy is better; it asked where each one wins. A cached monolith is cheap only while the cache is warm — **1,800** a call when it is, **22,500** when it is not — and progressive's **4,100** sits between the two, so the crossover is a hit rate near **89%**. The day desk lives far above that line; the night desk, eight calls well over an hour apart against a five-minute TTL, lives at zero. Monolithic and cached by day, progressive by night: about **9,392,800** token-equivalents a day, the lowest of the four plans. The slogan 'always cache' loses at the night desk; the slogan 'always send the map' loses by nearly twelve million.*

4 **Answer B**

The tell its 8 calls a night typically arrive 70–90 minutes apart

B ✓ A warm cached read is **18,000 × 0.1 = 1,800**; a cold write is **18,000 × 1.25 = 22,500**. The day desk stays warm, the night desk never does.

A **TR** Reverses the crossover and lands on the worst cell at both desks: about **21,500,000** a day against **9,392,800**.

C **VG** Confuses tokens sent with tokens billed; warm reads bill at **0.1×**, so the day desk pays **4,100** instead of **1,800** a call.

D **CM** Ignores the night spacing: **70–90**-minute gaps outlast a **5**-minute TTL, so every night call writes at **1.25×** = **22,500**.

↻ Revisit: D3 Loading Dock · obj 3.8 · Atlas A7 ; Dock Walk Station 8 ; Dock Gym station 3.8

TRAINED BY MISS FAMILY, NOT BY TOPIC

The Error Gym

12 items · 25 minutes · bar ≥ 10

Open this when three or more ledger rows share a family. Each item is built so that its most tempting wrong option belongs to one family:
OP · Over-permissive — the tempting option is over-permissive — and once, the narrowest option is too narrow to do the job.
OE · Over-engineered — the tempting option is over-built — and in exactly two of the four, the bigger option is right.
VG · Vocabulary gap — the trap is a word with two meanings: token, index, prompt, cache, scope, authN/authZ, mean/p95.
The items are mixed so you cannot predict the set. The key names each item's set.

1 Narwhal Shipping's port-rules assistant sends a 30,000-token rulebook ahead of every question, followed by the user's question. The rulebook changes once a quarter. p95 latency is 5.2 s against a 3-second target, and the API bill shows no cache reads at all. Asked about it, the platform engineer says caching is already on: the rulebook file is served from the company CDN with a 24-hour browser cache and a 98% hit rate. What should the team change?

- **A** Raise the CDN cache lifetime from 24 hours to the full quarter, since the rulebook only changes quarterly and the hit rate on the file would then approach 100%
- **B** Set a prompt-cache breakpoint at the end of the rulebook and keep that prefix byte-identical between calls, so that repeat requests read it from the API's cache
- **C** Move the assistant to the smallest model tier to bring p95 under 3 seconds, accepting some drop in answer quality on the harder rulebook questions
- **D** Split the rulebook into chunks in a vector index and retrieve only the relevant sections for each question, cutting the tokens sent on every call

2 Narwhal Shipping's booking assistant already writes structured logs, end-to-end traces with a trace ID on every hop, and a latency histogram for each stage. Its SLA promises a p95 response time of 3 seconds. Last month the p95 sat above 3 seconds for two days before a customer complained; the data showed the breach the whole time, but nobody was looking at it. The platform lead proposes a new observability platform with a 40-panel real-time dashboard. What should the team add?

- **A** A new observability platform with a 40-panel real-time dashboard, giving the whole team one shared view of latency, tokens, cost and errors per stage
- **B** A weekly review of the p95 charts in which the team looks closely at each stage and agrees owners for anything that seems slow
- **C** One alert on the p95 of the latency histogram at the 3-second SLA, sustained for 15 minutes and paged to the on-call engineer with a runbook link
- **D** An alert on mean response time above 3 seconds, paged to the on-call engineer, which is steadier and less noisy than a percentile alert

3 Saranda Mining's shift supervisors use a chat client connected to the company's MCP server, which already exposes the tools `get_equipment_status` and `log_incident`. Supervisors want a standard shift-handover report: at the end of a shift they pick 'Shift handover' from the client's menu, type the shift number, and get a filled template. The developer's ticket simply reads 'add the handover prompt'. How should the handover be added?

- **A** As a new section of the agent's system prompt holding the handover template, so that the model has the format on every call and can produce the report whenever a supervisor asks
- **B** As a separate handover agent that the chat client reaches over agent-to-agent messaging, so the report logic is owned and versioned by one team
- **C** As an MCP tool, `write_handover`, with a sharp description so the model calls it whenever it judges from the conversation that a shift has ended
- **D** As an MCP prompt on the server, `shift_handover`, taking the shift number as an argument, so that it appears in the client's menu for supervisors to choose

---

4 Saranda Mining's maintenance assistant searches 3,000 equipment manuals through one embedding index. Technicians ask two kinds of question: half paste an exact fault code or part number such as `E-417` or `SKF-23152`, and half describe a symptom in their own words. A month of logs shows the right chunk reaches the top 20 for 93% of symptom questions but only 58% of code questions. The answer SLA is 4 s p95 and the assistant runs at 3.1 s today. Which change should the team make?

- **A** Raise top-k from 20 to 150 and pass all 150 chunks to the model, which makes it far more likely that the chunk with the code is included
- **B** Add a BM25 keyword index beside the embedding index and fuse the scores, so that exact codes match on keywords while symptom questions keep their semantic matches
- **C** Replace the embedding index with a keyword index, since exact codes are where retrieval fails and keyword search matches them character for character
- **D** Add a query-planning agent that rewrites each failed question up to five times and searches the embedding index again until a chunk containing the code comes back

---

5 A security review of Wren Analytics' reporting agent lists Finding 7: 'The agent's token is far broader than its job.' The agent writes monthly reports for one client at a time, yet it queries the data warehouse with a single service token that can read all 40 clients' schemas. The developer's fix lowers `max_tokens` from 4,096 to 1,024 and trims the rows returned to the model. What should the fix have been?

- **A** A system-prompt rule naming the current client and forbidding queries on any other schema, with the agent's queries sampled monthly by the review team
- **B** The developer's change, extended with a lower context limit and a summary step, so that far fewer tokens about any client ever reach the model
- **C** A human analyst who approves every SQL query before it runs, giving each report a reviewer who knows all 40 clients' data
- **D** A short-lived warehouse credential issued per report run and scoped to that one client's schema, with the scope checked by the query tool before any SQL runs

---

6 Saranda Mining is building a RAG assistant over 9,000 site documents: blasting plans, equipment manuals and incident reports. Incident reports contain workers' medical notes, which only site safety officers may read. The prototype embeds every document into one vector index through a single service account, and the system prompt tells the model not to quote medical notes to other users. In testing, a contractor's question about a conveyor incident pulled a medical note into the retrieved context. What should change in the pipeline?

- **A** Run a redaction filter in code over each generated answer, removing medical terms before the reply is shown to anyone outside the safety team
- **B** Tag every chunk with its access group when it is indexed, and filter retrieval in code by the signed-in user's groups before any chunk reaches the prompt
- **C** Build a separate vector index, retriever and assistant for every role on site, so that no single index ever holds two roles' documents
- **D** Keep the single index and service account, and strengthen the system prompt with a firm rule and three refusal examples covering medical notes, which the red team can test before launch

---

7 Kutch Ceramics wants an assistant over its 1,400-page kiln operations manual, about 700,000 tokens, heading-structured and revised every month. Step 3 of the project plan reads 'index the manual'. The contractor delivered a lookup built from the manual's printed back-of-book index: each term maps to page numbers, and the assistant loads the matching pages. Questions in operators' own words, such as 'glaze pulling away from the edge', find nothing, because the manual calls it 'crawling'. What should step 3 have built?

- **A** No index at all: send the whole manual in the prompt on every call and cache it, which removes retrieval misses entirely
- **B** A richer back-of-book index with synonyms and operator slang added under every term, so that 'pulling away from the edge' also points to the crawling pages
- **C** A reranker and a query-planning agent on top of the existing page lookup, to reorder and retry the pages it returns until one matches
- **D** Heading-aware chunks with a little overlap, embedded into a vector index beside a BM25 keyword index, with changed sections re-indexed each month

---

8 Kutch Ceramics built an MCP server over its glaze-recipe database. Thirty designers use it from Claude Code and from the company chat app; today each runs a copy over stdio, with the database password in a local environment file. The team is moving to one shared remote server. Recipes for two unreleased product lines must be visible only to the six designers on those lines. Which setup should the team choose for the remote server?

- **A** Stdio kept on each laptop, with the database password moved into each designer's operating-system keychain so the plain-text secret is off the disk
- **B** Streamable HTTP with per-designer OAuth sign-in, the token verified in code on every request, giving each designer their own identity and an audit trail of recipes read
- **C** Streamable HTTP with per-designer OAuth sign-in, and the server checking each caller's product-line access in code before it returns any recipe
- **D** Streamable HTTP with one API key in the checked-in `.mcp.json`, so all thirty designers connect with no setup and the key is rotated in one place

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9 Wren Analytics runs a contract-clause assistant for its legal team. The SLA is 4 s p95, and the assistant runs at 2.3 s today. The accuracy bar is that the governing clause appears among the five chunks handed to the model for at least 90% of a 200-question test set; today it is 79%. A trial reranker over the top 150 candidates lifted that to 93% and added 0.7 s to p95. Which configuration decision should the team make?

- **A** Add the reranker together with a query-rewriting agent and a second vector index, pushing the hit rate as high as the trial budget allows
- **B** Leave the reranker out to protect the 2.3-second response time, and turn on streaming so that lawyers see the first words of each answer sooner
- **C** Move answer generation to the largest model tier, so that it reads the five retrieved chunks more carefully and picks out the governing clause
- **D** Add the reranker: the hit rate rises from 79% to 93% and p95 moves from 2.3 s to about 3.0 s, still inside the 4-second SLA

---

10 Narwhal Shipping is wiring its customer-service agent to the booking platform's API. Until now the agent only answered 'where is my container?' questions. The new brief adds one task: when a customer asks, the agent moves that booking's delivery window and confirms the new slot. The platform offers OAuth scopes `bookings:read`, `bookings:amend_window`, `bookings:cancel` and `bookings:admin`. Every chat is tied to one signed-in customer account. Which credential set should the agent's booking tool run with?

- **A** A short-lived token per chat holding `bookings:read` and `bookings:amend_window`, with a code check that each booking belongs to the signed-in customer
- **B** A short-lived token per chat holding `bookings:read` only, the narrowest scope on offer, with window-change requests answered by a link to the self-service portal
- **C** One `bookings:admin` service token for the agent, which covers status, window changes and any task a later brief adds, with every call written to the audit log
- **D** A short-lived token per chat holding `bookings:read` and `bookings:amend_window`, issued only after a fresh multi-factor check in code proves the customer owns the account

---

11 After Arjun's bloat audit cut KAVI's tool schemas from 2,314 to 811 tokens per call, everything KAVI sends ahead of the conversation — the trimmed tools, the system prompt and a short escalation guide — is a small, stable set of about 4,500 tokens, and every call uses all of it. A consultant proposes the API's tool search tool with `defer_loading: true` plus a summary catalogue for the guide, 'the same design that worked for the SOP library'. What should Arjun do with this set?

(A) Adopt the tool search tool with `defer_loading: true` and a summary catalogue for the guide, matching the SOP design and ready for tools added later

(B) Send the whole set on every call with a cache breakpoint at its end, keeping it byte-identical between calls so that each request reads it from the cache

(C) Chunk the escalation guide into the SOP vector index and retrieve only the relevant section on each call, trimming a few hundred tokens per request

(D) Keep sending the set as it is and review the token bill each month, moving to the catalogue only if the bill keeps rising

---

12 Wren Analytics in Bristol runs an orchestrator agent with eleven tools, among them `search_reports`, `read_report`, `email_client`, `publish_dashboard` and `delete_draft`. It is adding a subagent whose only job is to read past client reports and hand the orchestrator a 300-word summary of earlier findings. In the pilot the subagent stalled twice on scanned reports it could not open, and a developer suggests giving it 'everything the orchestrator has, so it never gets stuck'. Which tool set should the summary subagent get?

(A) `search_reports` and `read_report` only, with any report that will not open skipped quietly so that the summary still reaches the orchestrator on time

(B) `search_reports` and `read_report`, plus a supervisor agent that watches the summariser and restarts it each time it stalls on a report

(C) `search_reports` and `read_report` only, with an unreadable report returned to the orchestrator as a named error that lists the file it could not open

(D) All eleven of the orchestrator's tools, so that the subagent can work round any blocked report by itself and the pilot's two stalls do not recur

---

NO OPTIONS · FIND THE CLAUSE THAT DECIDES

The tell hunt

8 stems · 15 minutes · bar ≥ 7

A tell is the few words in a stem that separate the two best-looking answers. Here there are no options to lean on: underline the tell, write the answer's shape in one line, and name the family a careless reader would fall into. Answers are in the keys.

1 Narwhal Shipping's vessel-operations agent carries 18 tools, and their schemas cost about 3,900 tokens on every call. A clean-up ticket proposes removing every tool called fewer than five times in the last 90 days. That would drop seven tools: two near-identical berth lookups, an old weather feed, three retired reporting tools and `alert_duty_officer`, the agent's only route for handing a reefer temperature alarm to a person, called three times. The team lead wants the cut done before Friday's release. Which tools should go?

Underline the tell. The answer's shape: _____ The bait family: _____

2 Kutch Ceramics wants a Claude agent that compares supplier quotes each morning and drafts purchase orders for a buyer to approve. The supplier portal already has an MCP server that finance uses, and connecting the agent takes one line of config with finance's token, which never expires and also opens `release_payment` along with the quote and order tools. The developer adds that the agent's system prompt will tell it never to pay anyone. How should the agent connect?

Underline the tell. The answer's shape: _____ The bait family: _____

3 Wren Analytics codes open-text survey answers into 40 themes for its clients. The job runs on a mid-tier model, agrees with senior coders 94% of the time against a 92% bar, and costs more than the finance team likes. Each week's answers arrive by Friday 18:00, and the coded file goes out on Monday at 08:00. The finance lead suggests the smallest model tier, which scored 89% on the same 500-answer test. What should the team change?

Underline the tell. The answer's shape: _____ The bait family: _____

4 Narwhal Shipping wants to know why some container-status lookups are slow. Each lookup already writes a trace whose attributes include the container number, and a latency histogram labelled by pipeline stage. The platform engineer proposes adding `container_id` as a label on that histogram so a dashboard can show p95 per container. The tracking service sees about 1.2 million distinct containers a month, and the hosted metrics service bills per active time series. What should the team do?

Underline the tell. The answer's shape: _____ The bait family: _____

5 Saranda Mining's ventilation assistant answers questions from its gas-safety manual, which is split into fixed 400-token chunks. Prose questions go well, but 'what is the carbon monoxide limit at the level-6 decline?' returns confident numbers from the wrong column. Each gas-limit table runs to about 900 tokens, and only the first of its chunks carries the header row naming the columns. A colleague proposes adding a reranker and moving to a larger model. What should change?

Underline the tell. The answer's shape: _____ The bait family: _____

6 Wren Analytics' methods assistant answers analysts' questions from 6,000 internal methodology notes, retrieving the top 20 chunks by embedding similarity. The team lead reads 40 failed answers, says the model is being fed noise, calls it a precision problem and proposes cutting top-k to 5. The same review shows that in 31 of the 40, the note that answers the question was not among the 20 retrieved chunks at all. What should the team fix first?

Underline the tell. The answer's shape: _____ The bait family: _____

7 Saranda Mining's planning team wants the day's iron-ore benchmark price in its morning cost report. A developer proposes wrapping the price provider's REST API in an MCP server so that 'any agent can use it later'. The report is built by the team's own scheduled Python job, which calls the same endpoint once at 06:00, and no model decides whether or when the call is made. The provider's API is stable and versioned. Which integration mechanism fits?

Underline the tell. The answer's shape: _____ The bait family: _____

8 Kutch Ceramics' operations assistant connects to 12 MCP servers exposing 160 tools, whose schemas add up to about 95,000 tokens. The architect is choosing between sending every schema on every call behind a cache breakpoint and giving the model a searchable catalogue. Usage logs show that a typical request touches two or three tools, and the servers add or change tools most weeks. Which context strategy should the team choose?

Underline the tell. The answer's shape: _____ The bait family: _____

DOMAIN 3'S SHARE OF A REAL FORM

The Form Slice

12 items · 22 minutes · bar ≥ 10 inside the time

Sit this in the last two days before Mixed Mock II, with a timer. It has the exam's mix: 10 single-answer items and 2 multiple-response, spread across the eight objectives in roughly exam proportion, three of them leaning on other domains.

1 Tamarind Foods gives its store managers a Claude assistant and has already chosen a routing workflow: a small model classifies each question before it is handled. Logs show three kinds of question: 'what is the use-by date of batch TF-88213?', 'how many chilled deliveries arrived above 5 °C last week?' and 'what must I do if a customer reports an allergic reaction?'. Batch records and delivery readings sit in ERP tables; the food-safety manual is 300 heading-structured pages. Which three routes should the classifier send them to? Select THREE.

- **A** Counting questions go to hybrid retrieval over the exported delivery logs, with a reranker added so that the warmest deliveries rank first
- **B** Manual questions go to hybrid retrieval over heading-aware chunks of the food-safety manual, fusing keyword and embedding scores
- **C** Batch questions go to semantic search over embeddings of exported batch records, since embeddings also capture codes like TF-88213 found in the text
- **D** Counting questions go to a SQL-style aggregate over the delivery-readings table, and the model only phrases the number that comes back
- **E** Batch questions go to a structured lookup on the ERP batch table by exact batch ID, with no retrieval step in the path

2 Tamarind Foods' store assistant now handles about 400,000 calls a day across 900 stores in three stages: a classifier, retrieval and the answer. Finance, tuning cost, wants to know which stage drives the token bill. The metrics backend slowed to a crawl after an engineer added `order_id` as a label on the latency histogram, and storing a trace for every call would cost more than the assistant itself. Which two monitoring changes should the architect make? Select TWO.

- **A** Record tokens and cost as metrics per stage, labelled only by stage, model tier and region, and move `order_id` into logs and traces
- **B** Build a finance dashboard showing the total daily token bill, refreshed every hour from the provider's usage export for the whole assistant
- **C** Sample traces at a small fixed rate for normal calls, but keep the trace of every call that ends in an error or a refusal
- **D** Sample 1% of all traces uniformly, errors included, which cuts trace storage by 99% and treats every stage and every store alike
- **E** Keep `order_id` as a label but give the metrics backend more memory and longer retention, so that per-order latency stays on the dashboard

3 Magpie Media is building a Claude assistant for its licensing desk over 260 syndication contracts. Each contract is heading-structured, and every territory section ends with a royalty-rate table that averages about 900 tokens. The pilot cuts every document into fixed 400-token chunks with no overlap, and answers often quote a plausible rate under the wrong territory, because the rows that matter arrive without the table's header. Contracts change only when a deal is renegotiated. Which chunking change should the architect make?

- **A** Keep fixed-size chunking but raise the size to 1,000 tokens with 200 tokens of overlap, which holds an average 900-token table and its header in one piece
- **B** Chunk along the contract headings with a small overlap, and keep each royalty table whole, header row included, inside its territory section's chunk
- **C** Prepend a one-line summary of the whole contract to each 400-token chunk before embedding, as contextual retrieval suggests for missing context
- **D** Keep the 400-token chunks and add a reranker after vector search, so that the chunk holding the right territory's rows rises into the top results

4 Magpie Media's newsroom agents reach the CMS through an internal MCP server over Streamable HTTP. Every client already signs in with its own short-lived OAuth token, and the server logs which client made each call. A security review finds that the interns' summarising bot, authenticated correctly, successfully called `unpublish_article` on a live story. The CMS team wants the one change that would have refused that call. What should the architect recommend?

- **A** Route every CMS tool call from every client through an editor's approval queue, so that each action is checked by a person before it reaches the server
- **B** Move the summarising bot onto its own service account, its credentials verified in code on every call, so its calls cannot be mistaken for the desk agent's
- **C** Add a firm rule to the summarising bot's system prompt that it may read stories but may not change their publication status
- **D** Attach scopes to each client's token and make the server check, in code before `unpublish_article` runs, that the caller holds the publishing scope

5 Magpie Media's archive assistant searches 1.2 million clip records. Producers search in two ways: by exact clearance codes such as 'CLR-7Q-0419', and by descriptions such as 'aerial shot of a fishing harbour at dawn' that rarely match the catalogue's wording. Today's BM25 index finds codes reliably but misses most descriptive searches. A pilot with embeddings alone found the harbour clips but returned near-miss codes such as 'CLR-7Q-0491'. Which retrieval design should the architect adopt?

- **A** Run BM25 and embedding search side by side, fuse the two score lists into one ranking, and rerank the merged shortlist before use
- **B** Keep the BM25 index and add a synonym list curated by the archive team, so that descriptive words map onto the catalogue's own vocabulary
- **C** Keep embeddings alone but move to a larger embedding model, which should learn to separate close codes like CLR-7Q-0419 and CLR-7Q-0491
- **D** Let an agent try BM25 first, judge the hits, and retry with embeddings when they look weak, allowing up to five attempts per search

6 Puffin Ferries' booking agent in Aberdeen carries 14 tool schemas, about 3,100 tokens, on each of 50,000 calls a day. Thirty days of usage logs show four tools on 96% of calls; two of them, `get_booking` and `find_booking`, do the same job and the model often picks the wrong one. `raise_medical_alert`, which pages the duty officer when a passenger reports a medical emergency at sea, was called 11 times. The other nine tools were never called. Which tool set should the architect give the agent?

- **A** Keep only the four tools used on 96% of calls, which the thirty days of usage logs show covers nearly all traffic at the lowest token cost per call
- **B** Keep all 14 tools but cache the schema block, which cuts the repeated 3,100 tokens to a small fraction of their cost on every call
- **C** Merge the two booking lookups into one sharply described tool, keep the other two busy tools and `raise_medical_alert`, and drop the nine unused
- **D** Split the 14 tools across bookings, payments and safety subagents, each with its own schemas, and let an orchestrator route between them

7 Puffin Ferries in Aberdeen wants its spare-parts inventory system usable from four Claude clients: engineers' Claude Code sessions, the fleet desk's desktop app, a scheduling agent built on the Agent SDK, and a port-staff assistant planned for spring. The inventory system sits in Puffin's data centre; the clients run on laptops and in the cloud. The team expects to add stock-transfer and reorder tools during the year and wants to build each tool once. Which integration mechanism should the architect choose?

- **A** An inventory agent with its own goals and tools that each of the four clients consults agent-to-agent whenever a question about stock arises
- **B** Direct calls from each client's own tool code to the inventory system's REST API, which keeps every client free of any extra protocol layer
- **C** One remote MCP server over the HTTP+SSE transport that exposes the inventory tools behind OAuth sign-in, so that each client holds one long-lived event stream
- **D** One remote MCP server over Streamable HTTP that exposes the inventory tools, with OAuth sign-in for each client, so that every client reuses the same tools

8 Puffin Ferries in Aberdeen runs a passenger-help assistant over 1,400 policy and timetable pages, using hybrid search (BM25 plus embeddings) and passing the top 20 chunks to the model. On 300 golden questions the right chunk appears in the top 150 candidates 97% of the time but in the top 20 only 78% of the time, and wrong answers cluster on those misses. The answer SLA still has about one second of headroom. What should the architect change?

- **A** Re-chunk the 1,400 pages along their headings and re-embed them, since most retrieval misses start with poor chunk boundaries
- **B** Add a line to the system prompt telling the model to say it does not know when the retrieved chunks do not answer the question
- **C** Pass all 150 candidates straight to the model, which lifts the chance of the right chunk being in the prompt from 78% to 97%
- **D** Keep the hybrid search but retrieve about 150 candidates, rerank them against the question, and pass only the reranked top 20 to the model

9 Magpie Media's subtitle-checking agent sends six tool schemas and a house style guide on each of 120,000 calls a day: about 3,600 tokens that change twice a year. Every call uses at least four of the tools and the style guide. The model's minimum cacheable prefix is 1,024 tokens. A consultant proposes progressive discovery, a catalogue plus the tool search tool with `defer_loading: true`, citing an 86× token saving at a logistics firm. Which context strategy should the architect choose?

- **A** Send the whole block on every call, tools first and then the style guide, with a cache breakpoint after it so that later calls read it at about 0.1×
- **B** Send the whole block on every call without caching, since each cache write costs 1.25× and the block would cost more than it does today
- **C** Split the six tools across two subagents of three each, so that each subagent's context carries only half of the schemas and the style guide
- **D** Adopt the catalogue: mark all six schemas `defer_loading: true` behind the tool search tool and fetch the style guide on demand, as the consultant's 86× figure suggests

10 Kaveri Freight's agent KAVI answers customers' consignment questions through a carrier-portal tool. Arjun Bose finds that the tool signs in with one long-lived portal token that can read every customer's consignments. Sunita, from compliance, proposes an output guardrail: a classifier that scans each reply and blocks any consignment number that does not belong to the customer in the conversation. Meera Nair asks for the change that closes the integration gap at its source. What should the architect recommend?

- **A** Pass the portal token and the customer's consignment number as tool arguments, so the tool's code ties every portal call to one consignment in the logs
- **B** Deploy the proposed output classifier on every reply, which catches another customer's consignment number before the reply reaches the customer
- **C** Rotate the portal token every week and keep it in a secrets manager, so that it stays out of the prompt, the code and the logs
- **D** Issue short-lived tokens scoped to the signed-in customer, and have the tool's code confirm each consignment belongs to that customer before it reads

11 Kaveri Freight's exception assistant promises a first response within 2 s at p95. The operations dashboard shows a mean latency of 310 ms across 40,000 shipments a day, and the floor reads it as comfortably green. A histogram Arjun Bose adds shows p95 at 2,900 ms. Each ticket passes through the router, a retrieval call, a TMS tool call and the model, and the team must now find which stage makes the slow tickets slow. What should the architect add first?

- **A** A per-stage latency metric labelled with each shipment ID, so that the stages of any slow ticket can be looked up and compared straight from the operations dashboard
- **B** An alert that pages the on-call engineer whenever mean latency rises above 2 s, matching the SLA figure that the floor has signed up to
- **C** End-to-end tracing that carries one trace ID through the router, retrieval, TMS call and model, with a timed span per stage, keeping every error
- **D** Debug-level logs of the full prompt and reply for every ticket, which give a complete written record of what happened on each call

12 Tamarind Foods' food-safety answers must return within 4 s at p95, and accuracy on the 200-question golden set must stay at or above 90%. Today p95 is 4.9 s at 91%. The team tested four single changes: streaming the reply (first token 0.8 s, p95 4.8 s, 91%); cutting top-k from 30 to 15 (p95 3.7 s, 90.6%); removing the reranker (p95 3.9 s, 84%); moving to the smallest model tier (p95 3.1 s, 86%). Which sentence belongs in the decision record?

(A) To meet the 4 s p95 SLA, we cut top-k from 30 to 15: p95 falls to 3.7 s and accuracy dips to 90.6%, still above the 90% bar

(B) To meet the 4 s p95 SLA, we remove the reranker: p95 falls to 3.9 s and the pipeline loses a whole stage to run and maintain

(C) To meet the 4 s p95 SLA, we move to the smallest model tier: p95 falls to 3.1 s, the fastest of all four, at a lower cost per call

(D) To meet the 4 s p95 SLA, we stream the reply: the first token arrives in 0.8 s and accuracy stays at today's 91% on the golden set

KEYS · OPEN ONLY AFTER YOU HAVE MARKED THE SHEET

BAYS FIRST, THEN THE ERROR GYM, THE TELL HUNT AND THE FORM SLICE

Keys

All bay answers at a glance

3.1·1 · warm **C**	3.1·2 · warm **C**	3.1·3 · exam **B**	3.1·4 · exam **C**	3.1·5 · exam **D+E**	3.1·6 · head **D**	3.1·7 · head **B**	3.2·1 · warm **A**
3.2·2 · warm **B**	3.2·3 · warm **B**	3.2·4 · exam **C**	3.2·5 · exam **C**	3.2·6 · exam **A+C**	3.2·7 · exam **B**	3.2·8 · exam **D**	3.2·9 · head **A**
3.2·10 · head **D**	3.3·1 · warm **B**	3.3·2 · warm **A**	3.3·3 · exam **D**	3.3·4 · exam **A+B**	3.3·5 · exam **C**	3.3·6 · head **B**	3.3·7 · head **B**
3.4·1 · warm **A**	3.4·2 · warm **D**	3.4·3 · exam **C**	3.4·4 · exam **C**	3.4·5 · exam **C+E**	3.4·6 · exam **B**	3.4·7 · exam **C**	3.4·8 · head **C**
3.4·9 · head **A**	3.4·10 · head **D**	3.5·1 · warm **A**	3.5·2 · warm **B**	3.5·3 · warm **A**	3.5·4 · exam **A**	3.5·5 · exam **B**	3.5·6 · exam **C**
3.5·7 · exam **B+D**	3.5·8 · head **A**	3.5·9 · head **B**	3.5·10 · head **D**	3.6·1 · warm **B**	3.6·2 · warm **C**	3.6·3 · warm **B**	3.6·4 · exam **A**
3.6·5 · exam **A**	3.6·6 · exam **D**	3.6·7 · exam **A+B**	3.6·8 · head **A**	3.6·9 · head **D**	3.6·10 · head **D**	3.7·1 · warm **C**	3.7·2 · warm **A**
3.7·3 · exam **A**	3.7·4 · exam **C+D**	3.7·5 · exam **C**	3.7·6 · head **A**	3.7·7 · head **B**	3.8·1 · warm **C**	3.8·2 · warm **A**	3.8·3 · exam **C**
3.8·4 · exam **A+E**	3.8·5 · exam **A**	3.8·6 · head **D**	3.8·7 · head **A**				

CAPABILITY BLOAT

Key · bay 3.1

1 **Answer C** warm-up # number decides

The tell their schemas total 900 tokens

C ✓ **3,100 − 900 = 2,200** tokens saved per call; **2,200 × 25,000** calls = **55,000,000** tokens a day.

A **CM** Treats the whole catalogue as saved; the four kept schemas, **900** tokens, still travel on every call.

B **VG** **900 × 25,000** is what the agent still pays after the trim, not what the trim saves.

D **VG** Schemas are part of the input on every call, not a one-time load; the **2,200** tokens recur **25,000** times a day.

↻ Revisit: D3 Loading Dock · obj 3.1 ; Dock Walk Station 5 ; Dock Gym station 3.1

2 **Answer C** warm-up

The tell returns the same list price from a nightly copy

C ✓ Two lookalikes returning the same thing are bloat and a wrong-wrench risk; one sharply described tool removes the choice and the stale path.

A **SF** Catches only the errors a dealer notices, after the wrong price has gone out; the rest pass silently.

B **OE** Adds a schema and an extra call to solve what deleting one lookalike solves.

D **SD** A prompt preference is a sign; the stale lookalike stays in the catalogue, costing tokens and wrong picks.

↻ Revisit: D3 Loading Dock · obj 3.1 ; Dock Walk Station 5 ; Dock Gym station 3.1

3 **Answer B** exam # number decides
The tell each time for a reefer-truck failure where a missed escalation spoils the catch on board
B ✓ Trim by usage AND consequence: **1,150** + **210** = **1,360** ≤ **1,500,** and the rare escalation guards a loss that cannot be undone.
A **SD** Swaps a working tool for a request in the prompt; nothing guarantees the shift lead is ever reached.
C **HL** Trims by usage alone and removes the human path exactly where the harm, a spoiled catch, is irreversible.
D **CM** Ignores the **1,500**-token cap and keeps five tools that were never called.
↻ Revisit: D3 Loading Dock · obj 3.1 ; Dock Walk Station 5 ; Dock Gym station 3.1

4 **Answer C** exam
The tell it reads shipment status and the customs SOP, then hands a draft back to KAVI
C ✓ A subagent gets only the tools its job needs; a drafting job needs read tools, and an unused refund door is pure risk and pure tokens.
A **CM** Ignores that the subagent's job differs; its tool set should follow its own task, not reuse another agent's.
B **OP** Hands a refund door to a drafting job and re-adds the **2,314**-token catalogue the trim removed; one lock elsewhere is no excuse.
D **SD** A prompt line is a sign, not a lock; the refund tool is still in the subagent's hands.
↻ Revisit: D3 Loading Dock · obj 3.1 ; Dock Walk Station 5 ; Dock Gym station 3.1

5 **Answer D + E** exam
The tell each for a patient describing urgent symptoms
D ✓ Zero use and zero consequence: their schemas cost tokens on every call and buy nothing.
E ✓ Lookalikes cause wrong-wrench picks; one sharply described tool removes the **6%** confusion and two schemas.
A **OE** Adds orchestration and coordination cost instead of removing the seven dead tools and the lookalikes.
B **SD** More prompt tokens on top of the bloat; the three lookalikes still compete on every call.
C **HL** Trims by usage alone; each of those **31** calls was an urgent patient who needed a human.
↻ Revisit: D3 Loading Dock · obj 3.1 ; Dock Walk Station 5 ; Dock Gym station 3.1

6 **Answer D** head-to-head ⇄ head-to-head
The tell `hold_container` freezes a container in the yard pending customs; `cancel_booking` voids the shipment booking and releases the berth slot
D ✓ They do different things with different reversibility, and both are used daily; the confusion lives in the descriptions, and the description is the trigger.
A **CM** Merging suits true duplicates; here it hides an irreversible cancel behind a parameter and keeps the ambiguity.
B **SD** 'Think carefully' adds no information; identical descriptions still give the model nothing to choose by.
C **OE** A second agent per decision adds cost and latency to fix what two sentences of description fix.
↻ Revisit: D3 Loading Dock · obj 3.1 ; Dock Walk Station 5 ; Dock Gym station 3.1

7 **Answer B** head-to-head ⇄ head-to-head · # number decides · ▲ the bigger option is right
The tell The agent also compiles the 06:00 replenishment report, which needs the full bin list.
B ✓ Adding the tool IS right: **120 × 9,000** = **1,080,000** tokens a day against **18,000 × 3,000** = **54,000,000** for the workaround, and the report keeps its list.
A **SD** The tool still returns all **18,000** tokens; a prompt cannot shorten a result that has already arrived.
C **CM** Breaks the **06:00** replenishment report the agent compiles, which needs the full bin list; a flat tool count is not the goal.
D **OE** Three extra schemas that no log asked for; one narrow tool answers the **3,000** daily lookups.
↻ Revisit: D3 Loading Dock · obj 3.1 ; Dock Walk Station 5 ; Dock Gym station 3.1

AUTHENTICATION & AUTHORIZATION

Key · bay 3.2

1 **Answer A** warm-up

The tell any valid token, including the one issued to the appointment-reminder bot, can call `export_patient_list`

A ✓ Identity is proven (authN); what the caller may do (authZ) is unchecked. A per-tool scope check in code closes it.

B **CM** It is a shared remote service; stdio changes who can reach it, not the missing per-tool permission check.

C **VG** The server already knows who is calling; stronger proof of identity still lets the reminder bot export.

D **VG** Confuses the auth token with LLM tokens; output size says nothing about who may call the tool.

↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

2 **Answer B** warm-up

The tell runs a small MCP server over stdio on her own laptop

B ✓ stdio is a local child process on the same machine; it acts as the signed-in user, with no network listener to authenticate.

A **VG** HTTP+SSE is the deprecated remote transport; stdio uses neither HTTP nor sessions.

C **VG** OAuth is the pattern for remote Streamable HTTP servers; a stdio server is a local process with nothing listening on the network.

D **OP** Puts a credential in the model's hands, and a local stdio server needs no key at all.

↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

3 **Answer B** warm-up # number decides

The tell the first fraudulent call with that key came 40 hours later

B ✓ A copied access token would have expired within the hour, long before hour **40**; a short token life caps the window a thief can use.

A **VG** Strength stops guessing, not theft; a stolen long key works exactly as well as a short one.

C **OP** Concentrates the risk: one leak opens every integration, and the key is still permanent.

D **CM** A **90**-day rotation leaves the copied key valid at hour **40**; the number in the stem decides.

↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

4 **Answer C** exam

The tell each booking tool takes an `api_key` argument that the model fills in

C ✓ The model should never see or choose credentials. Binding each key to its own tool in code ends both the mix-up and the leak.

A **OP** The model still receives a usable key and still picks which one; encryption does not take it out of the credential path.

B **SD** A sign, not a lock: the model still holds every key and still chooses which one to send.

D **OE** Three agents still hold keys in prompts; adds orchestration and leaves the leak path open.

↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

5 **Answer C** exam

The tell that token can read every customer's consignments

C ✓ The token opens all bays, so authorization must be checked per object, in code, against the authenticated customer, before the tool runs.

A **OE** A model checking a model after the data is already fetched; a deterministic check before the call is cheaper and certain.

B **VG** Strengthens authN; the customer is already signed in. The gap is what a signed-in customer may read (authZ).

D **SD** A sign, not a lock: a typed-in ID still reaches the all-bays token, and a crafted message can talk past the rule.

↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

6 **Answer A + C** exam

The tell every change must be traceable to the integration that made it, and a leaked credential must stop working within the hour

A ✓ Short-lived tokens meet 'stop working within the hour'; a leaked token dies on its own.

C ✓ Separate identities make every change traceable to one integration, and each can be revoked alone.

B **OE** Adds costly review to sensor feeds and syncs; it neither identifies callers nor expires a leaked key.

D **CM** Misses both goals: a leaked key lives up to a week, not an hour, and every change still traces to one shared identity.

E **SD** A label the model is asked to write is not an identity; the sync job and a thief write untagged.

↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

7 **Answer B** exam · autopsy

The tell the only control is a system-prompt line stating the 15% limit

B ✓ Authorization in code, before the tool runs: the chat service's credential cannot grant **40%**, whatever text reaches the prompt.

A **SD** Still a sign, not a lock; the next forged note will not look like the three examples.

C **OP** Widens the exact door that was abused; any pasted note now opens it.

D **VG** Proves who the dealer is (authN); the dealer was genuine. The gap is what the chat service may grant (authZ).

↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

8 **Answer D** exam # number decides

The tell shows 37 calls refused with HTTP 403 for a missing `records:export` scope

D ✓ An audit log must record denials as well as grants; **37** refusals with **0** logged means the attempts that matter most are invisible.

A **OE** A mirror of a log that drops denials also drops them; one write of the refusal fixes it.

B **VG** A counter says how many, not who tried what; auditors need a per-event log entry, not a metric.

C **SF** Treats a log that silently drops refusals as proof; the error counter shows **37** attempts the log never saw.

↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

9 **Answer A** head-to-head ⇄ head-to-head

The tell the server accepts its single bearer token for every tool it exposes

A ✓ The token is the door: whoever holds it holds every tool. Only a server-side scope check stops a leaked bot token from rebooking.

B **SD** A sign, not a lock; a customer can talk the bot past it, and the token stays all-powerful.

C **OE** Secure, but it duplicates a server and its upkeep to do what one scope check in code does.

D **OP** A client-side filter trims bloat, not power: the token still opens `rebook_vessel_slot` to whoever holds it.

↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

10 **Answer D** head-to-head ⇄ head-to-head · ▲ the bigger option is right

The tell it will now serve 12 front-desk agents across Denver clinics from one hosted copy

D ✓ A shared remote server needs real authentication and per-tool authorization; adding the OAuth layer IS right here.

A **OP** One permanent shared key: no per-client identity for a scope check to tell apart, and one leak opens both tools everywhere.

B **CM** Ignores the one hosted copy; twelve local copies each need database access, and clerk logins carry no tool scopes.

C **VG** HTTP+SSE is deprecated; the current remote transport is Streamable HTTP with real authentication.

↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

ACCURACY-LATENCY TRADE-OFFS

Key · bay 3.3

1 **Answer B** warm-up

The tell naming the constraint, the dial that moved, the direction it moved and the number it cost or bought

B ✓ Constraint (**3 s** p95), dial (top-k), direction (**30** to **15**) and number (**0.4** points): all four parts in one sentence.

A **SD** Names a dial but no constraint and no number; 'no noticeable loss' is a feeling, not a figure.

C **CM** Borrows someone else's default; it names no constraint of Sutlej's and no cost or gain.

D **VG** Names the outcome metric as if it were the dial; no configuration setting is named, so nobody can repeat or reverse the change.

↻ Revisit: D3 Loading Dock · obj 3.3 ; Dock Walk Station 6 ; Dock Gym station 3.3

2 **Answer A** warm-up # number decides

The tell patients leave if no words appear within 1 s

A ✓ Streaming cuts time-to-first-token, not total time; the goal is first words within **1 s**, and **0.6 s** meets it.

B **TR** Spends accuracy that is only at the bar, and with no streaming the first words still arrive at **2.5 s**, past **1 s**.

C **VG** Streaming lowers time-to-first-token only; total generation stays about **7 s**.

D **TR** Batch is cheap and slow; most batches finish in under an hour, not in a second, the opposite of a **1 s** goal.

↻ Revisit: D3 Loading Dock · obj 3.3 ; Dock Walk Station 6 ; Dock Gym station 3.3

3 **Answer D** exam # number decides

The tell None of the three needs another's result.

D ✓ Independent calls can run at once: the wait becomes the slowest call, **1.5 s**, plus **1.4 s** of model time, **2.9 s**, with no change to the answer.

A **VG** Streaming cuts time-to-first-token; the full answer still takes **5.0 s**, past the **4 s** SLA.

B **TR** Spends the fixed **95%** accuracy and still misses: **4.4 s** is over the **4 s** SLA.

C **CM** Meets **4 s** by guessing a field the answer needs; accuracy cannot hold at **95%** on an inferred status.

↻ Revisit: D3 Loading Dock · obj 3.3 ; Dock Walk Station 6 ; Dock Gym station 3.3

4 **Answer A + B** exam

The tell must answer with p95 ≤ 4 s and at least 92% on its golden set

A ✓ Names the constraint, the dial, the direction and what it bought and cost; a complete justification.

B ✓ Constraint, dial, direction and number for both sides of the trade; the bar of **92%** still holds at **92.6%**.

C **VG** Answers a p95 constraint with a mean, which says nothing about the tail, and hides the **0.5**-point accuracy cost.

D **SD** No constraint and no number; nobody can check or reverse this decision from the sentence.

E **CM** Names the dials but not the constraint that moved them, nor what the moves cost.

↻ Revisit: D3 Loading Dock · obj 3.3 ; Dock Walk Station 6 ; Dock Gym station 3.3

5 **Answer C** exam # number decides · ▲ the bigger option is right

The tell The acceptance bar is ≥ 95% on the golden set with p95 ≤ 6 s.

C ✓ Only the bigger tiers clear **95%**, so the constraint decides; of those two, the mid-tier is cheaper and faster. Moving up IS right.

A **OE** Both bigger tiers clear the bar; the tie-break takes the cheaper, faster one. One extra point buys nothing the bar asks for.

B **VG** Streaming changes when text starts arriving, not accuracy; **91.0%** still fails the bar.

D **CM** Ignores the deciding clause: **91.0%** fails the **95%** bar, and wrong routes delay patients.

↻ Revisit: D3 Loading Dock · obj 3.3 ; Dock Walk Station 6 ; Dock Gym station 3.3

6 **Answer B** head-to-head ⇄ head-to-head · # number decides
The tell The service contract pays nothing for accuracy above the bar.
B ✓ When both clear the constraint, take the cheaper, faster option; the contract rewards nothing above **92%**.
A **TR** Spends latency and tokens chasing accuracy above the bar, eating the **2 s** headroom for nothing.
C **OE** Buys four times the cost for accuracy the contract does not pay for; more model than the bar needs.
D **VG** Streaming cuts time-to-first-token, not cost; Q is still four times the price for an unrewarded gain.
↻ Revisit: D3 Loading Dock · obj 3.3 ; Dock Walk Station 6 ; Dock Gym station 3.3

7 **Answer B** head-to-head ⇄ head-to-head · # number decides
The tell which can act only on the complete object
B ✓ The only tested change that clears both numbers for the complete object: **2.8 s** ≤ **3 s** and **97.6%** ≥ **97%**.
A **TR** Batch trades speed for cost; most batches finish in under an hour, the opposite of a **3 s** SLA.
C **VG** The ERP needs the whole object; streaming cuts time-to-first-token, and the complete verdict still takes **4.2 s**.
D **CM** Ignores the **97%** bar; **93.0%** fails it, so the fastest option does not clear the constraint.
↻ Revisit: D3 Loading Dock · obj 3.3 ; Dock Walk Station 6 ; Dock Gym station 3.3

OBSERVABILITY AT SCALE
Key · bay 3.4

1 **Answer A** warm-up # number decides
The tell Taking p95 as the 19th of the 20 sorted values
A ✓ The 19th of **20** sorted values is **3.8 s**, which is over **3 s**. The SLA is a promise about the tail.
B **VG** **4.4 s** is the maximum (the 20th value), not the 19th; right verdict, wrong percentile.
C **VG** **18** of **20** is the 90th percentile; p95 needs **19** of **20,** which is **3.8 s**.
D **VG** **29.3** ÷ **20** ≈ **1.5 s** is the mean; the SLA names p95, and two slow quotes hide inside a healthy average.
↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

2 **Answer D** warm-up
The tell each writes its own logs with no field they share
D ✓ A trace with a shared ID shows where the time went for ONE request; it is exactly the missing join across hops.
A **VG** A dashboard renders signal; it does not create the missing join, and timestamps blur when requests overlap.
B **VG** Metrics summarise many requests; one **9 s** claim barely moves a percentile and cannot be followed across hops.
C **SD** Process with no join key: five teams still cannot tell which of their lines belong to that claim.
↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

3 **Answer C** exam # number decides
The tell The yard handles about 90,000 containers a month
C ✓ **90,000 × 5 × 4** ≈ **1,800,000** series a month: the cardinality trap. Unbounded IDs belong in logs and traces, not metric labels.
A **VG** A hash is still one value per container; the series count, not the label length, is the problem.
B **SF** Still one series per container, and sampling skews the counts the dashboards and alerts rely on.
D **OE** Pays to store a label design that grows every month; logs and traces already answer per-container questions.
↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

4 **Answer C** exam # number decides
The tell p95 is 2,900 ms
C ✓ The mean hid the tail and the trace showed the stage; percentiles per stage plus trace links turn that one-off find into standing signal.
A **SD** A process with no metric, threshold or trace; it records complaints, not where the time went.
B **VG** A mean of **310 ms** already hides a **2,900 ms** p95; a stricter mean threshold still watches the wrong number.
D **OE** About **5%** of **40,000** calls is roughly **2,000** pages a day; alert fatigue, not a monitoring strategy.
↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

5 **Answer C + E** exam # number decides
The tell the storage budget, which allows about 60,000 traces a day, and about 0.4% of requests end in an error
C ✓ Metrics from full traffic keep dashboards and alerts accurate whatever the trace sampler keeps.
E ✓ **0.4%** of **1,200,000** = **4,800** error traces, all kept; about **55,200** sampled successes fill the **60,000** budget.
A **VG** A total-latency log line says how long, not where; the per-stage spans of failed requests are lost.
B **CM** Even one day of full traces is **1,200,000,** twenty times the **60,000**-a-day budget.
D **SF** Keeps about **240** of the **4,800** error traces; most failures vanish without a trace to open.
↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

6 **Answer B** exam · autopsy
The tell The only usage metric is total tokens a day for the whole pipeline
B ✓ Tokens and cost per stage show which stage grew; stage and model are bounded labels, so the metric stays cheap.
A **VG** Gives a metric label a trace's job: unbounded IDs as labels is the cardinality trap; per-claim detail belongs in logs and traces.
C **SD** No metric, number or gate; owners cannot see their stage's usage today, so there is nothing to watch.
D **TR** Spends the draft quality the stem fixes, truncating drafts and checks, before anyone knows which stage grew.
↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

7 **Answer C** exam ▲ the bigger option is right
The tell at random times and with many requests in flight
C ✓ The trace IS right here: overlapping requests make timestamp matching guesswork, and only a shared ID shows where one request's time went.
A **VG** With many requests in flight, lines from different requests interleave; a timestamp is not a trace ID.
B **CM** Acts on a guess: with no shared field across hops, nothing yet shows the model, rather than a customs API, is the slow hop.
D **SD** An alert says that a request was slow, not where; the engineer still has no way to follow it across hops.
↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

8 **Answer C** head-to-head ⇄ head-to-head · # number decides
The tell a genuine breach of the ETA SLA, p95 ≤ 3 s
C ✓ Few, actionable alerts on the promise itself: the p95 breach would have paged, and the noise moves to dashboards.
A **VG** Fewer alerts, but on the mean; the SLA is a p95, and a healthy mean can hide the breach that was missed.
B **SD** A diligence request with no change to the signal; alert fatigue returns within a week.
D **OE** Doubles the people reading noise; the SLA breach is still one alert among **140** that nobody acts on.
↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

9 **Answer A** head-to-head ⇄ head-to-head
The tell traces are kept for every request slower than 5 s
A ✓ A trace answers where the time went for ONE request, and one slower than **5 s** is guaranteed to be kept.
B **CM** Ignores that its trace is already kept; a replay may not reproduce a production delay at all.
C **VG** Percentiles summarise many requests; one **14 s** referral barely moves a p99, which is why the panels look green.
D **OE** Another aggregate panel; it still cannot show one request, and the answer is already stored.
↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

10 **Answer D** head-to-head ⇄ head-to-head · # number decides
The tell 99% of order confirmations each month arrive within 5 s
D ✓ **99%** within **5 s** is a p99 promise; p99 = **6.4 s** breaks it, and the alert must watch the tail the contract names.
A **VG** Reads p95 for p99; the contract names **99%**, and the 99th percentile was **6.4 s**.
B **OE** Five alerts where the contract names one percentile; extra pages breed alert fatigue.
C **VG** A mean is further from the tail than p95; it would have stayed green too.
↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

RAG PIPELINE: CHUNKING & INDEXING

Key · bay 3.5

1 **Answer A** warm-up # number decides
The tell add contextual embeddings, contextual BM25 and a reranker
A ✓ The published ladder: **5.7%** → **3.7%** (contextual embeddings) → **2.9%** (+ contextual BM25) → **1.9%** (+ reranking), a **67%** cut with all three.
B **VG** Reads cumulative cuts as stage-by-stage ones; **35**/49/67% are each measured from the **5.7%** baseline, ending at **1.9%**.
C **VG** A reranker rescores a broad pool (~**150**) and picks the **20,** so it does change what arrives: **2.9%** → **1.9%**.
D **VG** Treats keyword and embedding as the same signal; BM25 adds exact-term matching and takes failures to **2.9%**.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A1 ; Dock Walk Station 1 ; Dock Gym station 3.5

2 **Answer B** warm-up # number decides
The tell about 90,000 tokens in total
B ✓ Under ~**200k** tokens, skip RAG: the **90,000**-token guide fits in the prompt, and twice-yearly revisions leave the cached prefix stable.
A **OE** Chunking, an index and a reranker for a **90,000**-token corpus that fits in the prompt.
C **CM** Misreads 'twice a year': the prefix is stable for months between revisions, so a cache is rarely invalidated.
D **VG** Confuses the index with the chunking choice; it is still RAG, with **38** coarse vectors and a retrieval step for nothing.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A1 ; Dock Walk Station 1 ; Dock Gym station 3.5

3 **Answer A** warm-up
The tell the re-filing procedure had been cut across two chunks
A ✓ Chunk along meaning: heading-aware chunks keep the re-filing procedure whole; a little overlap carries context across boundaries.
B **SD** A sign, not a fix: the chunking still cuts the procedure, and the model cannot see what it was not given.
C **VG** Index vs chunk: better vectors of a half-procedure are still a half-procedure.
D **CM** Ignores the headings; a bigger fixed window still cuts at arbitrary points and adds unrelated text to every hit.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A2 ; Dock Walk Station 1 ; Dock Gym station 3.5

4 **Answer A** exam ▲ the bigger option is right
The tell the chunk text never says which SOP or which procedure it belongs to
A ✓ The chunk lacks its identity; a prepended context ('This chunk is from SOP **214** §**4,** customs e-way bill re-filing…') fixes both the vector and the keyword match.
B **CM** Dispatchers describe the situation, not SOP numbers; the filter needs a number nobody types.
C **SD** A sign, not a fix: the wrong chunk is still retrieved, and its text carries no SOP number to confirm.
D **VG** Index vs chunk: the chunk text lacks the SOP's identity; better vectors of the same words cannot add it.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A1 ; Dock Walk Station 1 ; Dock Gym station 3.5

5 **Answer B** exam · autopsy
The tell revised sheets are simply appended
B ✓ Weekly change needs a weekly re-index with version labels and stale versions dropped, so only the current figure can be retrieved.
A **SD** A sign, not a fix: both versions still reach the context, and nothing in either chunk tells the model which is current.
C **VG** A reranker scores relevance, not recency; two near-identical versions score alike.
D **CM** Ignores the weekly revision rate; stale chunks keep winning for up to three months.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A1 ; Dock Walk Station 1 ; Dock Gym station 3.5

6 **Answer C** exam
The tell the chunk holding that row does not hold the table's header row
C ✓ A row means nothing without its header; table-aware chunking keeps the column names with every row the model sees.
A **VG** Index vs chunk: the header is missing from the chunk text, and no embedding can restore it.
B **CM** Overlap is local; for row **30** of a **40**-row table the header is hundreds of tokens away.
D **TR** Spends tokens and precision on a gamble; nothing pairs the header with the right row.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A2 ; Dock Walk Station 1 ; Dock Gym station 3.5

7 **Answer B + D** exam
The tell many paragraphs switch between Kannada and English mid-sentence
B ✓ Mixed questions and mixed text need one index whose vectors place both languages in the same space; per-language indexes miss cross-language matches.
D ✓ Chunk along meaning, not language: splitting at every switch cuts steps in half.
A **VG** Embedding vs keyword: BM25 cannot match a Kannada paraphrase to English wording.
C **CM** Ignores that the SOPs are heading-structured; fixed windows cut procedures at arbitrary points.
E **OE** Adds a translation stage the problem does not need and throws away the wording staff actually search with.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A2 ; Dock Walk Station 1 ; Dock Gym station 3.5

8 **Answer A** head-to-head ⇄ head-to-head · # number decides
The tell 2,100 documents, about 640,000 tokens
A ✓ Skip RAG only under ~**200k** tokens; at **640,000** every call would read the whole library, cached or not.
B **OE** Pays for both designs; the full prefix makes the retrieval stage pointless.
C **VG** Reads ~**200k** as a per-prefix size limit; it is a corpus-size rule, and all **640,000** tokens still ride every call.
D **CM** Misses the ~**200k** line; a cache makes reads cheaper, but each of **3,000** daily calls still reads all **640,000** tokens.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A1 ; Dock Walk Station 1 ; Dock Gym station 3.5

9 **Answer B** head-to-head ⇄ head-to-head
The tell with no headings, numbered steps or section breaks
B ✓ With no structure to follow, overlap is the tool that carries meaning across a boundary.
A **TR** Buys recall with precision: every hit drags up to **2,000** tokens of unrelated observations into context.
C **CM** There are no headings or sections to follow in these reports.
D **VG** Contextual retrieval adds identity, not the missing half; the measurement still sits in the other chunk.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A2 ; Dock Walk Station 1 ; Dock Gym station 3.5

10 **Answer D** head-to-head ⇄ head-to-head · # number decides · ▲ the bigger option is right
The tell Each answer may carry at most 20 chunks
D ✓ The right chunk is in the pool but ranked low; reranking ~**150** → top **20** lifts it without breaking the cap (**2.9%** → **1.9%** in the published ladder).
A **SD** Watches the failure without fixing it; the right chunk still never reaches the model.
B **CM** Already in place; the stem says the pipeline uses contextual embeddings with contextual BM25.
C **TR** Breaks the **20**-chunk cap the bill depends on and floods the context with **130** weak chunks.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A1 ; Dock Walk Station 1 ; Dock Gym station 3.5

RETRIEVAL STRATEGY
Key · bay 3.6

1 **Answer B** warm-up
The tell what is the current status of work order WO-58213?
B ✓ An ID question has one right row; a structured query returns the live record, while RAG returns passages that merely mention the ID.
A **OE** Three sources and an agent loop for a question one query answers.
C **CM** Ignores 'current status': a day-old passage is still not the live record.
D **VG** Keyword search vs a structured lookup: BM25 still returns report passages, not the live record.
↻ Revisit: D3 Loading Dock · obj 3.6 · Atlas A3 ; Dock Walk Station 2 ; Dock Gym station 3.6

2 **Answer C** warm-up # number decides
The tell only 2 of those 20 chunks were about the question at all
C ✓ Recall = the right chunk made it in (**46**/50 = **92%**); precision = how much of the set is relevant (**2**/20 = **10%**). Precision is the weak one.
A **TR** Buys more recall when precision is the problem; **30** more chunks bring more look-alikes.
B **VG** Swaps the terms: **18** irrelevant chunks in **20** is a precision measure; the right chunk was present.
D **CM** Ignores the **2**-in-20 figure; the model is being handed **18** look-alikes per answer.
↻ Revisit: D3 Loading Dock · obj 3.6 · Atlas A3 ; Dock Walk Station 2 ; Dock Gym station 3.6

3 **Answer B** warm-up
The tell while the manuals say 'door gasket frost build-up'
B ✓ Keyword alone fails on paraphrase; embeddings match meaning, and fusing keeps BM25's strength on exact terms.
A **CM** Narrows where to look but not how to match; the wording gap remains inside the right manual.
C **SD** Moves the burden to people with no gate or check; the search still cannot match a description.
D **VG** A chunking fix for a matching problem: whole sections still share no words with the technician's description.
↻ Revisit: D3 Loading Dock · obj 3.6 · Atlas A3 ; Dock Walk Station 2 ; Dock Gym station 3.6

4 **Answer A** exam # number decides
The tell the total value of complimentary room upgrades at each of our 14 properties
A ✓ A sum is a question about rows, not passages; RAG returns a sample of chunks and cannot add up a table.
B **TR** Spends tokens and still asks the model to add up whatever sample arrives; nothing guarantees all **4,200** lines are retrieved.
C **SD** Process words with no lock; the totals are still guessed from fragments.
D **VG** A chunking change for an aggregation problem; retrieval still returns **20** rows, not all of them.
↻ Revisit: D3 Loading Dock · obj 3.6 · Atlas A3 ; Dock Walk Station 2 ; Dock Gym station 3.6

5 **Answer A** exam · autopsy
The tell each chunk carries a status label (draft, effective, withdrawn) and an effective date
A ✓ The labels already exist; a filter keeps drafts and withdrawn versions out of the context, so they cannot poison the answer.
B **SD** A sign with no lock: the draft still reaches the context and is still trusted.
C **OE** Three indexes and a comparison step, and drafts still reach the context; one filter does the job.
D **VG** A reranker scores relevance, not approval status: the draft matched the question best, so it stays first.
↻ Revisit: D3 Loading Dock · obj 3.6 · Atlas A3 ; Dock Walk Station 2 ; Dock Gym station 3.6

6 **Answer D** exam ▲ the bigger option is right
The tell Descriptions are answered well; exact references often return the neighbouring rule
D ✓ Keyword matching catches exact references; embeddings keep the descriptions; fusion plus reranking serves both patterns.
A **VG** Embedding vs keyword: near-identical codes are a keyword job; any embedding places them close together.
B **CM** Fixes references and breaks descriptions; keyword alone fails on paraphrase.
C **OE** Two agents and two indexes where one hybrid retriever serves both patterns.
↻ Revisit: D3 Loading Dock · obj 3.6 · Atlas A3 ; Dock Walk Station 2 ; Dock Gym station 3.6

7 **Answer A + B** exam
The tell 'where is KF-2481 right now?', 'how many customs holds did we clear this week?'
A ✓ An ID question has one live row; a structured query answers it exactly and fast.
B ✓ A count is a question about rows; RAG cannot count, an aggregate can.
C **TR** Spends tokens and latency and still counts from a sample of passages.
D **VG** Keyword search vs structured lookup: BM25 returns notes that mention KF-2481, not where it is now.
E **CM** A how-to lives in SOP prose (SOP **214** §**4**); there is no record to look up. Policy questions go to hybrid RAG.
↻ Revisit: D3 Loading Dock · obj 3.6 · Atlas A3 ; Dock Walk Station 2 ; Dock Gym station 3.6

8 **Answer A** head-to-head ⇄ head-to-head
The tell Can booking HH-30917 be refunded under our monsoon cancellation policy?
A ✓ The question has two shapes: an ID part (structured lookup) and a policy part (hybrid RAG). Each part goes to its own route.
B **CM** Serves the ID half and drops the policy half; the refund depends on the festival-date exceptions.
C **OE** Two known routes do the job; an open-ended agent with every system adds cost and risk for a fixed path.
D **VG** Keyword match vs structured lookup: an indexed copy of a booking is a passage, not the live record.
↻ Revisit: D3 Loading Dock · obj 3.6 · Atlas A3 ; Dock Walk Station 2 ; Dock Gym station 3.6

9 **Answer D** head-to-head ⇄ head-to-head · # number decides
The tell type consignment IDs such as KF-2481 into the assistant to get the latest status
D ✓ An ID lookup is not a retrieval problem; the structured route is **₹4** against **₹1,800** and meets the ≤ **2 s** SLA.
A **CM** Ignores 'latest status': a cached answer goes stale as the shipment moves.
B **OE** Another stage on the wrong route; it adds cost and latency to a lookup that needs no retrieval.
C **VG** Keyword-leaning is right for codes inside documents; an ID lookup wants the live record, not a better-ranked note.
↻ Revisit: D3 Loading Dock · obj 3.6 · Atlas A3 ; Dock Walk Station 2 ; Dock Gym station 3.6

10 **Answer D** head-to-head ⇄ head-to-head
The tell Fitters rarely name their depot in a question, but the app always knows which depot a fitter is signed in at
D ✓ The depot is known from the session, so a filter makes the wrong depot's steps impossible to retrieve.
A **TR** Adds eight wrong versions to the context when precision is the problem.
B **SD** A sign, not a lock: nothing stops Depot **7**'s chunk being retrieved and trusted.
C **CM** Helps only when the question names the depot; fitters rarely do, so Depot **7**'s chunk still matches as well.
↻ Revisit: D3 Loading Dock · obj 3.6 · Atlas A3 ; Dock Walk Station 2 ; Dock Gym station 3.6

CONNECTION PROTOCOLS
Key · bay 3.7

1 **Answer C** warm-up
The tell No model reads or decides anything in this job
C ✓ One fixed, stable call from your own code with no model choice is the case for a direct API call.
A **CM** MCP pays off with many clients or model-chosen tools; here one script makes one call and no other app uses the data.
B **VG** Agent-to-agent is for an autonomous agent with its own goals; the board offers a plain API.
D **OE** Adds a model and a loop to a deterministic call that has not changed in three years.
↻ Revisit: D3 Loading Dock · obj 3.7 · Atlas A4 ; Dock Walk Station 7 ; Dock Gym station 3.7

2 **Answer A** warm-up ↔ leans on D7
The tell Claude Code to run `get_batch_record` without asking
A ✓ MCP tools are named `mcp__<server>__<tool>`; this rule allows exactly the read-only tool and leaves the other two asking.
B **VG** MCP tools are matched by the full `mcp__lims__get_batch_record` name; the bare tool name matches nothing.
C **VG** Drops the `mcp__` prefix; the rule names a tool that does not exist, so Claude Code keeps asking.
D **OP** Allows all three tools, including the two that change data.
↻ Revisit: D3 Loading Dock · obj 3.7 · Atlas A4 ; Dock Walk Station 7 ; Dock Gym station 3.7

3 **Answer A** exam # number decides · ▲ the bigger option is right
The tell 24 integrations must be patched whenever the property-management API changes
A ✓ Many clients sharing the same tools is where MCP pays: write the server once, and every MCP client, Desktop and Code included, uses it.
B **VG** The vendor offers an API, not an autonomous agent with goals of its own.
C **OE** Merges four products into one agent to fix what is a connector problem.
D **CM** Claude Desktop and Claude Code cannot import a library, and four redeploys remain; a server written once serves every client.
↻ Revisit: D3 Loading Dock · obj 3.7 · Atlas A4 ; Dock Walk Station 7 ; Dock Gym station 3.7

4 **Answer C + D** exam

The tell dispatchers pick the customs triage checklist themselves from a menu

C ✓ Prompts are user-invoked templates; the dispatcher, not the model, chooses the checklist.

D ✓ Tools are model-invoked actions; KAVI chooses when and with what to call these two.

A **VG** The MCP prompt primitive is not the system prompt; dispatchers choose this checklist themselves.

B **VG** Resource vs tool: the client attaches the SOP, so it is application-provided data, a resource (`sop://214`).

E **OE** One server can expose tools, resources and prompts; a second deployment adds a door for nothing.

↻ Revisit: D3 Loading Dock · obj 3.7 · Atlas A4 ; Dock Walk Station 7 ; Dock Gym station 3.7

5 **Answer C** exam # number decides

The tell against the current MCP specification (2026-07-28)

C ✓ The current spec's remote transport is Streamable HTTP and the protocol is stateless; remote callers need real authentication.

A **VG** stdio is the local, same-machine transport; **30** applications calling over a network need Streamable HTTP.

B **CM** HTTP+SSE is deprecated; a server built against the current spec uses Streamable HTTP.

D **OE** Builds infrastructure for state the stateless spec no longer keeps; each request carries its own `_meta`.

↻ Revisit: D3 Loading Dock · obj 3.7 · Atlas A4 ; Dock Walk Station 7 ; Dock Gym station 3.7

6 **Answer A** head-to-head ⇄ head-to-head

The tell through one copy running in the data centre

A ✓ One remote copy means Streamable HTTP; per-analyst OAuth gives the server an identity to check and record.

B **OP** One shared key identifies no analyst, so access cannot be checked or recorded per person.

C **VG** HTTP+SSE is the deprecated transport; the remote transport is Streamable HTTP.

D **CM** Ignores the single data-centre copy; **40** copies spread database credentials and nothing records access centrally.

↻ Revisit: D3 Loading Dock · obj 3.7 · Atlas A4 ; Dock Walk Station 7 ; Dock Gym station 3.7

7 **Answer B** head-to-head ⇄ head-to-head

The tell which owns its fleet rules and tools and decides by its own goals which paths it will give up

B ✓ The other side is an autonomous agent owning its goals and tools; that is the agent-to-agent case.

A **VG** MCP exposes tools to call; the partner offers an agent that decides for itself, and no functions to wrap.

C **OE** Builds a new shared system holding both fleets' keys to settle what is a negotiation.

D **CM** Ignores that the partner publishes no booking endpoints.

↻ Revisit: D3 Loading Dock · obj 3.7 · Atlas A4 ; Dock Walk Station 7 ; Dock Gym station 3.7

PROGRESSIVE VS MONOLITHIC CONTEXT

Key · bay 3.8

1 **Answer C** warm-up # number decides

The tell 9 tool stubs of 25 tokens each, one 260-token header and 20 SOP summaries of 180 tokens each

C ✓ **9 × 25 + 260 + 20 × 180 + 120 = 4,205**; **362,460 ÷ 4,205 ≈ 86**.

A **VG** Confuses a summary (**180** tokens) with a stub (**25** tokens).

B **CM** Drops the **120**-token fetch the stem says every call makes.

D **CM** Misreads 'one **260**-token header' as one per stub.

↻ Revisit: D3 Loading Dock · obj 3.8 · Atlas A7 ; Dock Walk Station 8 ; Dock Gym station 3.8

2 **Answer A** warm-up # number decides
The tell A 3,000-token block of tool schemas and system prompt
A ✓ Below the model's minimum, nothing is cached, silently; Haiku **4.5**'s minimum is **4,096** tokens.
B **VG** The cache reads the prefix in order tools → system → messages; tools and system cache first.
C **CM** Calls come every **5** seconds; and a re-write would show cache-creation tokens, not zero.
D **VG** Haiku **4.5** caches; its minimum prefix is **4,096** tokens, not zero support.
↻ Revisit: D3 Loading Dock · obj 3.8 · Atlas A7 ; Dock Walk Station 8 ; Dock Gym station 3.8

3 **Answer C** exam # number decides
The tell with cache reads billed at 0.1× of the input price
C ✓ **362,460 × 0.1 = 36,246,** about **8.6 × 4,205** (even at **0.05×**, **18,123** is **4.3×**); large, weekly-changing sets favour progressive discovery even against a cached monolith.
A **TR** Spends **8.6** times the context under a fixed **₹6** ceiling and calls it close.
B **VG** A cache read is cheaper, not free: **0.1×** of **362,460** is still **36,246**.
D **OE** Adds four agents and their coordination to dodge a problem a catalogue already solves.
↻ Revisit: D3 Loading Dock · obj 3.8 · Atlas A7 ; Dock Walk Station 8 ; Dock Gym station 3.8

4 **Answer A + E** exam
The tell an incident typically uses 4 or 5 of them, and the servers add or change tools most weeks
A ✓ Spent log results are the bulk of the context; server-side context editing clears them.
E ✓ A large, changing, rarely used set is the case for the API-native catalogue.
B **OE** Twelve subagents and an orchestrator for what a catalogue and context editing fix.
C **CM** Ignores that **4** or **5** of **140** tools are used and the set changes weekly; a large, changing, rarely used set wants a catalogue.
D **VG** Clears thinking, not tool results; the log dumps are what fill the context.
↻ Revisit: D3 Loading Dock · obj 3.8 · Atlas A7 ; Dock Walk Station 8 ; Dock Gym station 3.8

5 **Answer A** exam
The tell builds its tools array per request: a router picks the 6–10 of 45 tools it judges relevant
A ✓ The cache reads the prefix from the first token; a changing tools array breaks it. A fixed array with deferred tools keeps the prefix stable.
B **CM** Requests come every few seconds; expiry is not the cause, a changing prefix is.
C **VG** Breakpoint vs prefix: the prefix already differs at its first tokens, wherever the breakpoint sits.
D **CM** Ignores that the selection itself varies per request; **6–10** of **45** gives many different prefixes.
↻ Revisit: D3 Loading Dock · obj 3.8 · Atlas A7 ; Dock Walk Station 8 ; Dock Gym station 3.8

6 **Answer D** head-to-head ⇄ head-to-head · # number decides · ▲ the bigger option is right
The tell an average call uses 9 of the 12
D ✓ **9,600 × 0.1 = 960**; **500 + 9 × 800 = 7,700**. A small, stable, almost-always-used set is cheaper sent whole and cached.
A **OE** Adds a search step to nearly every call for a set that is almost always used.
B **VG** Confuses the write price with the read price; a warm cache bills reads at **0.1×**.
C **CM** Ignores that **9** of **12** schemas are fetched anyway: **7,200** more tokens every call.
↻ Revisit: D3 Loading Dock · obj 3.8 · Atlas A7 ; Dock Walk Station 8 ; Dock Gym station 3.8

7 **Answer A** head-to-head ⇄ head-to-head · # number decides
The tell needed until the session ends
A ✓ `exclude_tools` protects the one result needed all session; the spent log searches are still cleared.
B **VG** Clearing thinking does not change which tool results `clear_tool_uses` removes.
C **SD** A sign, not a setting: relies on the model copying **6,000** tokens of limits faithfully every turn.
D **CM** Ignores the **30** or more searches: ten later searches still push the record out, and **70,000** more tokens of spent logs ride along.
↻ Revisit: D3 Loading Dock · obj 3.8 · Atlas A7 ; Dock Walk Station 8 ; Dock Gym station 3.8

THE SET EACH ITEM TRAINS IS NAMED FIRST

Key · Error Gym

1 **Answer B** VG set · obj 3.3 ⇄ head-to-head
The tell the API bill shows no cache reads at all
B ✓ The CDN caches a file download; only prompt caching lets the model reuse the processed **30,000**-token prefix, cutting input cost and time on every repeat call.
A **VG** Same word, other world: a browser or CDN cache never reaches the model, so the bill still shows no cache reads.
C **CM** Ignores the zero cache reads that point at the real waste: the same **30,000** tokens processed afresh every call.
D **OE** A **30,000**-token, quarterly-changing rulebook is far under ~**200k** tokens: put it in the prompt and cache it instead.
↻ Revisit: D3 Loading Dock · obj 3.3 · Atlas A7 ; Dock Walk Station 6 ; Dock Gym station 3.3

2 **Answer C** OE set · obj 3.4 ⇄ head-to-head
The tell the data showed the breach the whole time, but nobody was looking at it
C ✓ The signal already existed; what was missing was something that speaks up. One SLA-tied, actionable alert closes exactly that gap.
A **OE** Dashboards render signal; they do not watch it. Nobody looking was the gap, and **40** panels add more to look at.
B **SD** A process with no threshold and a seven-day gap; it would have missed a two-day breach just as quietly.
D **VG** The SLA is a promise about the tail; a mean can sit well under **3** seconds while the p95 breaches.
↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

3 **Answer D** VG set · obj 3.7 ⇄ head-to-head
The tell they pick 'Shift handover' from the client's menu, type the shift number
D ✓ MCP prompts are user-invoked templates that clients show in a menu and fill from arguments; that is exactly what supervisors described.
A **VG** Reads 'prompt' as the system prompt; nothing appears in the menu, and the template costs tokens on every call.
B **OE** A template does not need an autonomous agent with its own goals; one server primitive does the job.
C **CM** Tools are model-invoked; supervisors asked to choose the handover themselves from the menu.
↻ Revisit: D3 Loading Dock · obj 3.7 · Atlas A4 ; Dock Walk Station 7 ; Dock Gym station 3.7

4 **Answer B** OE set · obj 3.6 ▲ the bigger option is right · # number decides
The tell half paste an exact fault code or part number
B ✓ A second index is the right extra stage here: semantic search fails on rare identifiers, keyword search catches them, and the fused scores keep the **93%** symptom path.
A **TR** Floods the context and spends the **0.9 s** of headroom; **150** is the shortlist to rerank, not the set to hand the model.
C **CM** Fixes the code half and breaks the symptom half: keyword search alone fails on paraphrase.
D **OE** An agent loop around the same embedding index; rewording a part number does not make a rare code rank, and each retry spends latency.
↻ Revisit: D3 Loading Dock · obj 3.6 · Atlas A3 ; Dock Walk Station 2 ; Dock Gym station 3.6

5 **Answer D** VG set · obj 3.2
The tell queries the data warehouse with a single service token that can read all 40 clients' schemas
D ✓ The finding is about the auth token: narrow it to the one client the run serves and enforce it in code, so other schemas are unreachable.
A **OP** Keeps the all-schemas token and trusts the prompt as the control; monthly sampling finds a leak weeks late.
B **VG** Treats the auth token as LLM tokens; the warehouse token can still read all **40** schemas.
C **OE** Adds a review step to every query instead of narrowing the key; the token still opens every schema.
↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

6 **Answer B** OP set · obj 3.5
The tell which only site safety officers may read
B ✓ The narrowest working key: one index still serves everyone, but each user retrieves only what their groups allow, so the note never enters a contractor's context.
A **SF** The note has already reached the model; paraphrases slip past a term filter and the leak goes unseen.
C **OE** Duplicates the whole pipeline per role when chunk metadata and one filtered index do the job.
D **OP** Trusts the prompt to guard a secret that is already inside the context window; a sign, not a lock.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A1 ; Dock Walk Station 1 ; Dock Gym station 3.5

7 **Answer D** VG set · obj 3.5 # number decides
The tell Questions in operators' own words, such as 'glaze pulling away from the edge', find nothing
D ✓ A retrieval index over meaning-sized chunks finds paraphrases through embeddings and exact terms through BM25, and monthly re-indexing keeps it current.
A **CM** About **700,000** tokens is far past the ~**200k** point below which skipping RAG pays; every call would carry the whole manual.
B **VG** Improves the book's index, not a retrieval index; every new phrasing needs a hand-added synonym.
C **OE** Reranking and retrying a term lookup cannot surface pages it never matched; the paraphrase still finds nothing.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A2 ; Dock Walk Station 1 ; Dock Gym station 3.5

8 **Answer C** OP set · obj 3.7 ⇄ head-to-head
The tell must be visible only to the six designers on those lines
C ✓ A remote transport needs real authentication; per-user identity plus an authorization check in code is the narrowest key that serves all thirty and hides the two lines.
A **CM** Ignores the move to one remote server; every laptop still holds a password that reads every recipe.
B **VG** AuthN without authZ: everyone is identified, yet all thirty can still read the unreleased lines.
D **OP** A shared key names nobody and opens every recipe to all thirty; keys stay personal, never in checked-in config.
↻ Revisit: D3 Loading Dock · obj 3.7 · Atlas A4 ; Dock Walk Station 7 ; Dock Gym station 3.7

9 **Answer D** OE set · obj 3.3 ▲ the bigger option is right · # number decides · ⇄ head-to-head
The tell A trial reranker over the top 150 candidates lifted that to 93% and added 0.7 s to p95
D ✓ Only one configuration clears the accuracy bar, and it does so while staying inside the latency SLA; the constraint decides, and the extra stage earns its place.
A **OE** Goes past the bar the reranker already clears, and the extra agent and index eat the SLA headroom untested.
B **CM** Guards latency that has **1.7 s** of headroom and ignores the failing bar; streaming cuts time to first token, not misses.
C **TR** Spends latency on generation, the stage that is not failing; no model can read a clause retrieval never handed it.
↻ Revisit: D3 Loading Dock · obj 3.3 · Atlas A1 ; Dock Walk Station 6 ; Dock Gym station 3.3

10 **Answer A** OP set · obj 3.2 ⇄ head-to-head
The tell Every chat is tied to one signed-in customer account.
A ✓ Two scopes are the smallest set that still does the new job, and the ownership check in code stops one customer moving another customer's booking.
B **CM** Narrowest, but it cannot do the job: the brief now has the agent itself move the delivery window.
C **OP** Admin also opens cancel and every customer's bookings; an audit log records misuse after the fact, it does not prevent it.
D **VG** A second identity check is authN; nothing stops a signed-in customer naming someone else's booking, which is an authZ gap.
↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

11 **Answer B** OE set · obj 3.8 # number decides
The tell a small, stable set of about 4,500 tokens, and every call uses all of it
B ✓ Small, stable and always used is where monolithic plus caching wins; every call reuses the cached **4,500** tokens at a fraction of the input price.
A **OE** Progressive discovery pays off for large, changing, rarely used sets; here it adds fetch turns for content every call needs.
C **CM** Ignores that every call uses all of it; retrieval adds a stage and can miss a part that is always needed.
D **SD** A review with no gate or number; it leaves the obvious win, caching the stable prefix, on the table.
↻ Revisit: D3 Loading Dock · obj 3.8 · Atlas A7 ; Dock Walk Station 8 ; Dock Gym station 3.8

12 **Answer C** OP set · obj 3.1 ⇄ head-to-head
The tell whose only job is to read past client reports
C ✓ A reader gets the reading tools and nothing that sends, publishes or deletes; the stall is fixed by reporting the failure upward, not by adding doors.
A **SF** Drops a signal: the orchestrator receives a confident summary with findings silently missing.
B **OE** Adds an agent and a loop for two stalls; a named error lets the orchestrator decide what to do.
D **OP** Hands email, publish and delete to a job that only reads; the stall was an error-handling gap, not a missing tool.
↻ Revisit: D3 Loading Dock · obj 3.1 · Atlas A5 ; Dock Walk Station 5 ; Dock Gym station 3.1

THE TELL, THE SHAPE, THE BAIT

Key · Tell hunt

1 · obj 3.1 · bait **HL · Human loop skipped**
Tell the agent's only route for handing a reefer temperature alarm to a person
Shape Trim by usage AND consequence: cut or merge the lookalike and dead tools, keep `alert_duty_officer`.

2 · obj 3.2 · bait **OP · Over-permissive**
Tell also opens `release_payment` along with the quote and order tools
Shape Its own short-lived credential scoped to quotes and draft orders, enforced in code; never finance's payment-capable token or a prompt rule.

3 · obj 3.3 · bait **TR · Trade-off reversed**
Tell the coded file goes out on Monday at 08:00
Shape Keep the mid-tier model and run the weekend job through the Message Batches API at half the price (**24**-hour expiry fits the **62**-hour window); latency is not the constraint.

4 · obj 3.4 · bait **CM · Constraint missed**
Tell about 1.2 million distinct containers a month
Shape Keep metric labels bounded (stage, route); find the slow containers by querying traces and logs filtered on container number.

5 · obj 3.5 · bait **OE · Over-engineered**
Tell only the first of its chunks carries the header row naming the columns
Shape Keep each table whole with its header row (or repeat the header in every piece); heading-aware chunks for the prose.

6 · obj 3.6 · bait **VG · Vocabulary gap**
Tell the note that answers the question was not among the 20 retrieved chunks at all
Shape A recall failure: widen and sharpen what gets retrieved (contextual chunks, hybrid BM25 + embeddings, rerank a wider pool); cutting top-k makes it worse.

7 · obj 3.7 · bait **OE · Over-engineered**
Tell no model decides whether or when the call is made
Shape A direct API call from the scheduled job; MCP pays off only with many clients, many tools or model-chosen calls.

8 · obj 3.8 · bait **CM · Constraint missed**
Tell a typical request touches two or three tools
Shape Progressive discovery: the tool search tool with `defer_loading: true`, loading only the few schemas a request needs.

SCORE IT AGAINST 22 MINUTES

Key · Form Slice

1 **Answer B + D + E** obj 3.6 · Retrieval strategy ↔ leans on D1
The tell Batch records and delivery readings sit in ERP tables
B ✓ A how-to question over prose is RAG's home ground; hybrid catches exact terms and paraphrased symptoms alike.
D ✓ A count is a question about rows; retrieval returns a few passages and cannot sum a table.
E ✓ An ID is a key into a table; a direct lookup is exact, fast and cheap, while retrieval adds cost and can return a near-miss record.
A **OE** Adds a stage that improves passage ranking but still cannot answer 'how many'; RAG cannot count rows.
C **VG** Embeddings capture meaning, not exact identifiers; a code like TF-88213 can return a near-miss batch. An ID is a key, not a meaning.
↻ Revisit: D3 Loading Dock · obj 3.6 · Atlas A3 ; Dock Walk Station 2 ; Dock Gym station 3.6

2 **Answer A + C** obj 3.4 · Observability at scale ↔ leans on D4
The tell wants to know which stage drives the token bill
A ✓ Answers finance per stage with bounded labels; the unbounded order ID goes where it belongs, in logs and traces.
C ✓ Sampling makes traces affordable at **400,000** calls a day; keeping **100%** of errors keeps the signal that matters.
B **CM** A total cannot say which stage drives the bill; dashboards render signal, they do not create per-stage signal.
D **SF** Throws away **99%** of the failures with the noise; errors must be kept in full.
E **OE** Buys infrastructure to carry an unbounded label; the fix is to remove the label, not to feed it.
↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

3 **Answer B** obj 3.5 · RAG pipeline: chunking & indexing # number decides · ⇄ head-to-head
The tell Each contract is heading-structured, and every territory section ends with a royalty-rate table
B ✓ Heading-aware chunks follow the document's meaning, and a table kept with its header tells the model which territory each rate belongs to; a little overlap carries context across boundaries.
A **CM** Still cuts on length: tables above the average, or starting mid-chunk, still lose their header. The cut must follow structure, not size.
C **VG** Contextual retrieval adds chunk-specific context; a contract-wide summary still does not say which territory a headerless block of rows belongs to.
D **OE** A reranker reorders chunks; it cannot restore a header the chunker cut away, so the rows stay unlabelled.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A2 ; Dock Walk Station 1 ; Dock Gym station 3.5

4 **Answer D** obj 3.2 · Authentication & authorization ⇄ head-to-head
The tell authenticated correctly, successfully called `unpublish_article` on a live story
D ✓ The bot proved who it was (authN), but nothing limited what it may do (authZ). A scope check in code before the tool runs refuses the call.
A **OE** Puts a human on thousands of harmless reads to stop one class of write; a scope check stops it deterministically.
B **VG** Clients already have their own identities; another account changes who it is, not what it may do. Authentication is not authorization.
C **SD** A sign, not a lock: the server still accepts the call whenever the model ignores the sentence or is talked past it.
↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

5 **Answer A** obj 3.6 · Retrieval strategy ⇄ head-to-head
The tell descriptions such as 'aerial shot of a fishing harbour at dawn' that rarely match the catalogue's wording
A ✓ Each method covers the other's blind spot: BM25 for exact codes, embeddings for paraphrase. Fused scores serve both query shapes in one search.
B **CM** Keyword search fails on paraphrase; a hand-kept list cannot keep up with descriptions that rarely match the catalogue's wording.
C **VG** Embeddings compare meaning, and a clearance code has none to compare; exact identifiers need keyword matching, not a bigger embedder.
D **OE** Adds a loop, latency and cost to choose between two searches that a single fused query runs together.
↻ Revisit: D3 Loading Dock · obj 3.6 · Atlas A3 ; Dock Walk Station 2 ; Dock Gym station 3.6

6 **Answer C** obj 3.1 · Capability bloat # number decides · ⇄ head-to-head
The tell pages the duty officer when a passenger reports a medical emergency at sea
C ✓ Trim by usage AND consequence: drop the unused nine, merge the lookalikes that cause wrong-wrench picks, keep the rare tool whose absence is costly.
A **HL** Drops the path to the duty officer for medical emergencies; a rarely used tool stays when missing it is costly. It also keeps the lookalikes.
B **CM** Caching lowers cost, but the lookalike tools still cause wrong picks; it ignores 'the model often picks the wrong one'.
D **OE** Adds agents and coordination to carry nine tools nobody calls; trimming and merging fixes it in one agent.
↻ Revisit: D3 Loading Dock · obj 3.1 · Atlas A4 ; Dock Walk Station 5 ; Dock Gym station 3.1

7 **Answer D** obj 3.7 · Connection protocols ⇄ head-to-head
The tell wants to build each tool once
D ✓ Many clients and a growing tool set are where MCP pays off: write the server once, any MCP client uses it; remote clients need Streamable HTTP with real authentication.
A **OE** Agent-to-agent suits a counterpart that owns its own goals; an inventory system needs tools, not a second autonomous agent.
B **CM** Right for one fixed call from one codebase; with four clients and more tools coming, every tool gets rebuilt four times.
C **VG** HTTP+SSE is the deprecated transport; current MCP is stdio for local and Streamable HTTP for remote.
↻ Revisit: D3 Loading Dock · obj 3.7 · Atlas A4 ; Dock Walk Station 7 ; Dock Gym station 3.7

8 **Answer D** obj 3.5 · RAG pipeline: chunking & indexing # number decides · ▲ the bigger option is right
The tell in the top 150 candidates 97% of the time but in the top 20 only 78% of the time
D ✓ The right chunk is found (**97%** in **150**) but ranked too low (**78%** in **20**); a reranker fixes the ordering and the headroom pays for it: ~**150** → rerank → top **20**.
A **CM** The right chunk is already retrieved **97%** of the time in the top **150**; chunking is not the fault, ranking is.
B **SD** Changes the wording of failures, not their rate; the right chunk is still missing from **22%** of prompts.
C **TR** Buys recall by flooding the prompt: precision collapses, retrieved tokens rise **7.5×** with latency close behind, and the model must still find the needle.
↻ Revisit: D3 Loading Dock · obj 3.5 · Atlas A1 ; Dock Walk Station 1 ; Dock Gym station 3.5

9 **Answer A** obj 3.8 · Progressive vs monolithic context # number decides · ⇄ head-to-head · ↔ leans on D2
The tell Every call uses at least four of the tools and the style guide
A ✓ Small, stable and always used: monolithic plus caching wins. **3,600** tokens clears the **1,024** minimum, so every later call reads the prefix cheaply.
B **VG** Confuses the one-off write premium with the price of every later read (**0.1×**); at **120,000** calls a day, caching is far cheaper.
C **OE** Adds agents and coordination to shrink a **3,600**-token block that caching already makes cheap.
D **CM** Progressive discovery wins for large, rarely used sets; here every call fetches most of it anyway, adding steps for no saving.
↻ Revisit: D3 Loading Dock · obj 3.8 · Atlas A7 ; Dock Walk Station 8 ; Dock Gym station 3.8

10 **Answer D** obj 3.2 · Authentication & authorization ↔ leans on D5 · ⇄ head-to-head
The tell the change that closes the integration gap at its source
D ✓ The gap is authorization at the door: an all-bays token. A per-customer scope checked in code before the read means other customers' data never enters the context.
A **OP** Hands the credential to the model and still leaves an all-bays token; the model must never see or choose credentials.
B **CM** A useful second lock, but downstream: the token still reads every bay and the data still enters the context. It ignores 'at its source'.
C **VG** Credential hygiene is authentication; a rotated all-bays token still opens every customer's consignments. Who it is is not what it may do.
↻ Revisit: D3 Loading Dock · obj 3.2 · Atlas A5 ; Dock Walk Station 3 ; Dock Gym station 3.2

11 **Answer C** obj 3.4 · Observability at scale # number decides · ⇄ head-to-head
The tell find which stage makes the slow tickets slow
C ✓ Metrics showed that the tail is slow; a trace shows where the time went for one request, stage by stage (one slow ticket's trace read **4,100 ms**).
A **CM** At **40,000** shipments a day, an unbounded label explodes the metric store; IDs belong in logs and traces, not metric labels.
B **VG** The SLA is about p95, not the mean; the mean sits at **310 ms** while p95 already breaches, so this alert never fires.
D **OE** Full prompt-and-reply logging adds heavy volume and cost, yet logs record events; they do not show where one request's time went.
↻ Revisit: D3 Loading Dock · obj 3.4 · Atlas A6 ; Dock Walk Station 4 ; Dock Gym station 3.4

12 **Answer A** obj 3.3 · Accuracy–latency trade-offs # number decides · ⇄ head-to-head

The tell accuracy on the 200-question golden set must stay at or above 90%

A ✓ Constraint + dial + direction + number. It is the only change that clears both the latency SLA and the accuracy bar, so the constraint decides.

B **TR** Clears latency by spending accuracy: **84%** falls well below the **90%** bar.

C **CM** Fastest and cheapest, but **86%** misses the **90%** accuracy bar; when only one option clears both, the constraint decides.

D **VG** Streaming cuts time-to-first-token, not total time; p95 is still **4.8 s**, so the SLA is still breached.

↻ Revisit: D3 Loading Dock · obj 3.3 · Atlas A1 ; Dock Walk Station 6 ; Dock Gym station 3.3

ONE ROW PER MISS · THE LEDGER, NOT THE PAGE ORDER, PICKS YOUR NEXT STATION

Miss ledger

Item	Obj	Family you chose	The tell you missed (copy it)	Re-sit on (48 h)	✓